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ENHANCING EFFICIENTNET-B0 USING A LOCAL-GLOBAL TRANSFORMER LAYER FOR DIABETIC RETINOPATHY CLASSIFICATION

A Thesis Proposal
Presented to the Faculty of the
College of Computer and Information Sciences
Polytechnic University of the Philippines
Sta. Mesa, Philippines

In Partial Fulfillment of the Requirements for the Degree
Bachelor of Science in Computer Science

by

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June 2025

DECLARATION ON THE USE OF ARTIFICIAL INTELLIGENCE

We are Alano, Bajit, Labuyo, Lague, and Vega, students at the Polytechnic University of the Philippines here declare that we HAVE used Artificial Intelligence to complete this research project/thesis/dissertation titled Enhancing Efficientnet-B0 Using a Local-Global Transformer Layer for Diabetic Retinopathy Classification

The use of Artificial Intelligence is in the following manner:

AI Tool Used	Detailed Prompts and AI Outputs	How the output was incorporated in the project
Gemini	Can you list statistical tests appropriate for comparing three multi-class models? Include references for each	Provided insights into how to choose the most appropriate statistical test to answer our hypothesis.
ChatGPT	Can you provide a simple outline of how we can better structure the introduction of our study?	The AI provided suggestions about how we can structure the flow of our introduction. Some were used to refine our existing writeup.
ChatGPT	I want to connect these paragraphs for better cohesion. Can you give me ideas for a transition?	Some of the transitional phrases that were suggested were used to improve the flow of the discussion.
ChatGPT	How can I diagram a conceptual framework using our independent and dependent variables?	AI suggested some structural ideas. One template was chosen and adapted by refining it to fit the research context.

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CERTIFICATION OF ORIGINALITY

This is to certify that the research work presented in this thesis, *ENHANCING EFFICIENTNET-B0 USING A LOCAL-GLOBAL TRANSFORMER LAYER FOR DIABETIC RETINOPATHY CLASSIFICATION* for the degree Bachelor of Science in Computer Science at the Polytechnic University of the Philippines embodies the result of original and scholarly work carried out by the undersigned. This thesis does not contain words or ideas taken from published sources or written works that have been accepted as basis for the award of the degree from any other higher education institution, except where proper referencing and acknowledgement of work were made.

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ABSTRACT

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Keywords:

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Chapter 1

THE PROBLEM AND ITS SETTING

Introduction

Diabetic Retinopathy (DR) is a progressive eye disease that remains as one of the leading causes of vision impairment. Specifically, it is a major complication of Diabetes (*Diabetes mellitus*) that is caused by long-term damage to the blood vessels of the retina due to prolonged high blood sugar levels. According to the Asia Pacific Tele-Ophthalmology Society (APTOS, 2019), it is the leading cause of blindness among working-aged adults. Likewise, The World Health Organization (WHO, 2023) lists it among the top global causes of vision loss.

In the Philippines, DR is also a growing public health concern. In 2021, an estimated 4.3 million Filipinos were diagnosed with diabetes, while another 2.8 million remained undiagnosed (Cando et al., 2023). DR, the most common complication of diabetes, has become a leading cause of preventable blindness, particularly in region 3. Despite the availability of effective treatments, it remains a major cause of vision loss among diabetic patients nationwide (Philippine Eye Research Institute [PERI], 2021).

The Philippine Eye Research Institute underscores the need for routine retinal screening to enable early detection. However, widespread implementation remains challenging due to geographic and systemic barriers. Furthermore, retinal image evaluation is a highly specialized task requiring trained experts, rigorous quality control, and continuous upskilling. The limited availability of such specialists in the country has made manual screening difficult to sustain at scale.

Clinically, DR advances through four main stages: mild, moderate, severe, and proliferative. In its early stages, the condition is often asymptomatic or presents only subtle

retinal changes. However, if left untreated or diagnosed late, DR can lead to irreversible damage, including retinal neovascularization, hemorrhages, and permanent blindness.

Given these, early detection of DR is important for effective treatment and vision preservation, as allowing it to progress further only makes it worse and untreatable (Prasad et al., 2023). However, one of the issues faced in this field is that manually diagnosing diabetic patients takes time for ophthalmologists. It is time-consuming, requires continuous quality control, and is prone to human error (Bhaskaranand et al., 2016). Moreover, in rural and low-resource areas where access to ophthalmologists and facilities are limited, this challenge is only further exemplified (APTOS, 2019; Cao et al., 2022).

It is these challenges that highlight the need for an automated diagnostic system that can assist ophthalmologists in the early detection of DR.

In recent years, many studies have explored the use of Computer Vision (CV) in various medical imaging tasks, with Convolutional Neural Networks (CNNs) having shown to have great performance over traditional Machine Learning models. Researchers have applied CNNs to areas such as radiology for detecting tumors, dermatology for classifying skin lesions, pathology for identifying cancerous cells in histopathology images, and cardiology for analyzing echocardiograms.

In the context of retinal-disease imaging, and DR classification specifically, CNNs have likewise shown promise. Numerous studies have reported encouraging results using various architectures to grade retinal images. However, most published models fall short of the reliability expected for clinical deployment as many report overall accuracies only in the mid-70% range.

Such shortcomings are concerning, as even small drops below the clinically accepted threshold could translate to thousands of potential misdiagnoses at population scale. Kim (2021) highlights this gap, noting that any DR classifier achieving less than 90% accuracy remains limited for medical use, because misclassification can carry serious consequences.

Despite extensive research on CNN-based approaches for DR classification, studies across various architectures, including MobileNet, EfficientNet-B0, Inception-ResNet-v2, and VGG-16 have reported accuracies ranging only from the low 60% to mid-80% range, even after applying transfer learning, data balancing techniques, or architecture-specific refinements. These results, as shown in works by Kim (2021), Ganesh et al. (2021), Haripriya et al. (2022), and others, indicate a consistent performance ceiling among CNN-based models. Even methodological studies aimed at improving preprocessing and sampling strategies acknowledge that while such techniques offer incremental improvements, they remain insufficient to meet clinical deployment standards (Huang et al., 2023).

DR classification, at its core, is a fine-grained visual recognition task that requires models to detect subtle details in retinal images to accurately differentiate between closely related stages of severity. However, CNNs, while effective for many general image classification tasks, exhibit notable limitations when applied to fine-grained problems due to how their architectures work inherently.

One issue lies in their use of pooling operations, which, while helpful for achieving spatial invariance, also discard high-resolution detail. Additionally, CNNs typically operate on local receptive fields. And although this structure is powerful, it inherently limits the model's ability to capture global context.

Furthermore, CNNs also tend to perform best when inter-class variance is high, such as in distinguishing entirely different object categories. In DR classification, however, the differences between classes are often very small (low inter-class variance), and individual variation among patients within a class can be quite high (high intra-class variance).

One way to address these limitations of CNNs in fine-grained classification is through the integration of attention mechanisms. These allow models to focus on the most informative parts of an image, enhancing their ability to capture subtle yet discriminative features. Among its various types, self-attention has demonstrated strong performance in many CV tasks due to its ability to model long-range dependencies. However, its downside lies in its computational demands (Hegde et al., 2023). This is particularly important in the context of DR classification, where models should ideally be lightweight for deployment without compromising performance (Aasrith et al., 2024).

As a result, more efficient alternatives such as the Squeeze-and-Excitation (SE) block and the Convolutional Block Attention Module (CBAM) have been more favored due to their relatively low overhead. Yet even these mechanisms have their own limitations. Zhou et al. (2023) highlighted that despite the improvements they bring, significant challenges remain. They emphasized that while they are favored in medical imaging for their reduced computational cost, their feature extraction abilities often lag behind heavier architectures.

These contexts provide an avenue where further research can be done.

In a comparative study by Ganesh et al. (2021), the authors noted that EfficientNets present a viable alternative to other state-of-the-art models as it offers faster convergence, reduced training time, and comparable accuracy, all while requiring lower

computational demands. These characteristics make it a promising candidate for further exploration.

In a different context, in a study by Babita and Nayak (2024), the authors introduced a novel Local-Global Spatially-aware Transformer Layer (LSTL) as part of their LiCT-Net architecture. Their key innovation was in replacing the traditional self-attention mechanism of Transformers with a more efficient attention alternative which integrates both global and local spatial dependencies while having reduced computational complexity.

Given the fine-grained feature extraction required in DR classification, the LSTL's ability to balance local precision and global context is especially promising. Furthermore, its lightweight design aligns with the needs of DR models, particularly in resource-constrained environments.

As such, it is in this context that our study is situated. In this paper, we propose to adapt the LSTL layer from LiCT-Net into an EfficientNet-B0 backbone for the task of DR classification. Our objective is to evaluate whether this integration can address the limitations of standard CNNs in capturing fine-grained details, while maintaining computational efficiency. Moreover, our experiments will test the viability of adapting LSTL for the EfficientNet-B0, a more compact CNN framework, for DR classification.

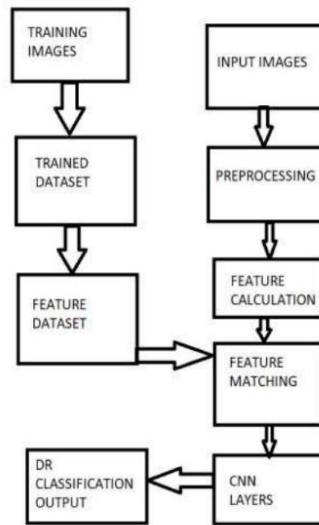
Theoretical Framework

Based on insights from previous studies, the research adopts EfficientNet-B0 as an optimized architecture for DR classification. Leveraging its proven balance of accuracy and computational efficiency, the framework builds upon earlier deep learning approaches to enhance diagnostic precision and explore further improvements in feature extraction and detection performance.

DR Classification

A study by Dayal et al. (2020) proposes a CNN-based approach for detecting DR by processing retinal images through a structured pipeline involving feature extraction, matching, and classification.

Figure 1. Block Diagram of CNN-based Diabetic Retinopathy Detection System (Dayal et al. 2020)



This figure illustrates the multi-stage architecture employed in DR detection process:

The system starts with a set of training images, which are used to construct a trained dataset. From this dataset, a feature dataset is generated by extracting meaningful attributes related to retinal abnormalities, such as microaneurysms or hemorrhages.

Simultaneously, new input images undergo preprocessing to enhance image quality and normalize illumination variations. This step is followed by feature calculation, where the relevant characteristics are computed to mirror those in the training set.

Next, feature matching is performed by comparing features from the input image with those in the pre-trained feature dataset. These matched features are passed into the CNN layers, where deep hierarchical learning enables complex pattern recognition.

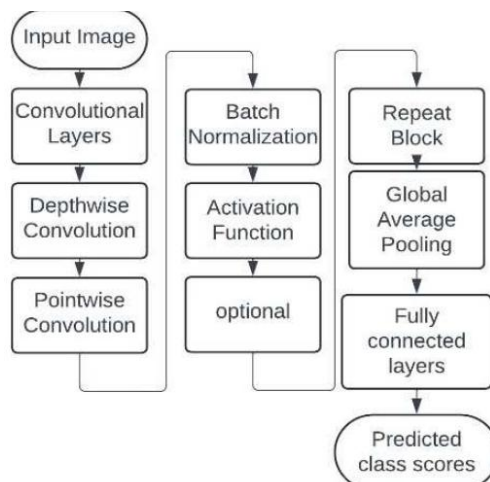
Finally, the network produces the DR classification output, categorizing retinal images based on DR severity. This output supports automated diagnosis, improving accuracy and reducing reliance on manual screening.

By combining feature engineering with CNN-based classification, the framework ensures a balance between traditional image analysis and modern deep learning. It streamlines early DR detection, offering a scalable tool for medical diagnosis.

EfficientNet-B0

EfficientNet-B0 serves as the backbone architecture of the deep learning model for DR classification. As a lightweight CNN, it was selected for its optimal trade-off between accuracy and computational efficiency, making it ideal for deployment on low-resource devices.

Figure 2. Flowchart of EfficientNet-B0 Architecture (Aasrith et al. 2024)



Aasrith et al. (2024) illustrates the EfficientNet-B0 architecture used in their study for DR classification, which follows a structured series of feature extraction and classification blocks:

The model begins with convolutional layers to extract low-level image features. These layers are followed by depthwise and pointwise convolutions, which together perform efficient spatial and channel-wise feature processing. Depthwise convolution applies a separate filter to each input channel, preserving spatial locality, while pointwise convolution (1x1 filters) integrates information across channels, significantly reducing the parameter count.

Subsequently, the model applies batch normalization to stabilize training and improve convergence, followed by non-linear activation functions to enhance feature discrimination. These operations are structured into repeatable blocks that allow for deeper hierarchical learning. After these blocks, global average pooling reduces each feature map to a single value per channel—minimizing model complexity and helping to prevent overfitting. The final output is passed through fully connected layers with a Softmax activation, yielding class probabilities for five DR severity levels.

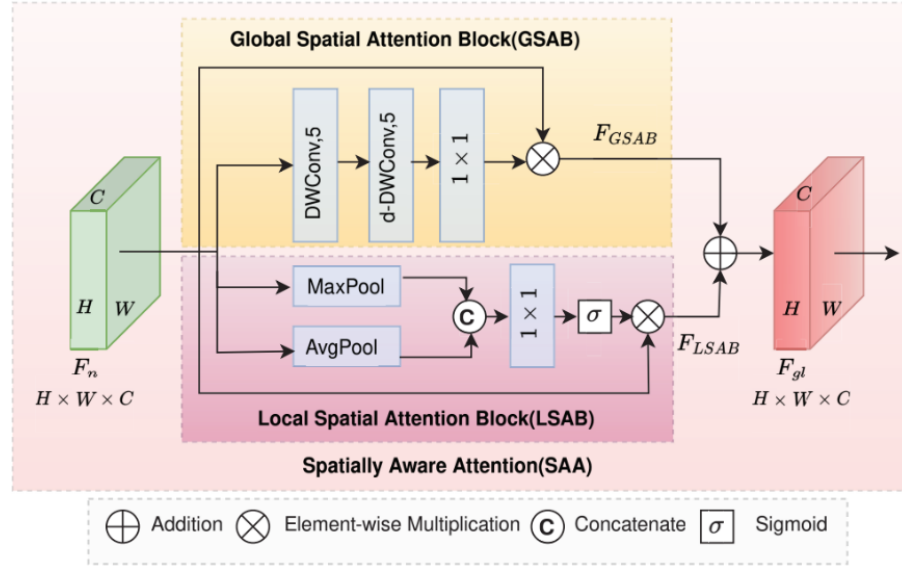
By leveraging this architecture, the model effectively captures both low-level and high-level features critical for accurate DR classification. Its efficiency and scalability make it particularly suitable for real-world medical applications with constrained computational resources.

LSTL

LSTL is a key component of LiCT-Net, designed to enhance feature extraction by capturing both local and global dependencies in histopathological

images. Traditional CNNs and vision transformers often struggle to balance broad contextual awareness with fine-grained local feature representation. LSTL addresses this challenge by integrating multiple attention-based components to refine spatial feature dependencies, making it particularly effective for medical image classification tasks (Babita & Nayak, 2024).

Figure 3. Illustrative representation of LSTL (Babita & Nayak, 2024)



This figure from the LiCT-Net paper illustrates the structure of LSTL, which consist of three key components: LayerNorm, SAA, and Feedforward Neural Network (FNN).

The LayerNorm module ensures stable activations across layers, preventing gradient instability and improving training efficiency. This normalization step is crucial for maintaining consistency in feature scaling, allowing the network to process diverse lesion patterns effectively. By standardizing feature distributions, LayerNorm helps prevent vanishing or exploding gradients, ensuring smoother optimization during training.

The SAA mechanism is divided into two-sub blocks: GSAB and LSAB. The GSAB expands the receptive field to capture broad contextual relationships, ensuring that the model can recognize complex lesion structures. Meanwhile, the LSAB focuses on fine-grained local dependencies, refining feature representations by emphasizing spatially relevant details. Together, these attention mechanisms enable LSTL to balance global and local feature extraction, improving classification accuracy. The attention feature map is computed by combining outputs from GSAB and LSAB, ensuring that both large-scale contextual dependencies and localized patterns are effectively captured.

Finally, the FNN processes the refined features, integrating local and global dependencies for final classification. The FNN consists of depthwise convolutions followed by pointwise convolutions, ensuring efficient feature transformation while maintaining computational efficiency. This structure allows the network to refine extracted features before passing them to the classification layer.

By leveraging LSTL, deep learning models can maintain global coherence while emphasizing fine-grained spatial features, which is particularly beneficial for medical imaging applications such as DR classification. This ensures that models remain efficient while accurately detecting relevant patterns.

Conceptual Framework

Figure 4. Conceptual Framework of the Study

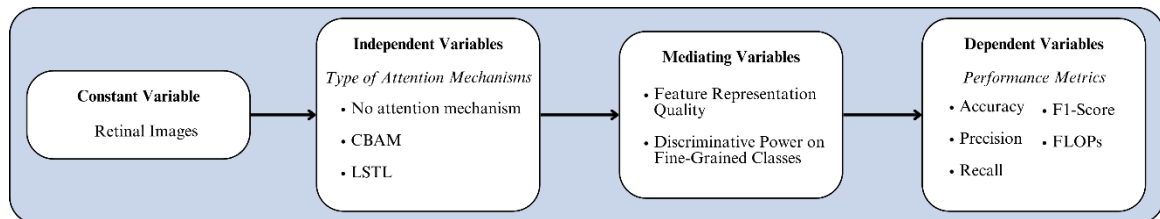


Figure 4 illustrates the conceptual framework of the study that defines the relationship among the variables in the study. At the core of the research is the investigation of how different attention mechanisms, when integrated into the EfficientNet-B0 model, influence the performance of DR classification.

The constant variable is the retinal fundus images used across all experiments. This ensures that each model configuration is trained and tested under the same dataset conditions, allowing for a controlled comparison.

The independent variables are the types of attention mechanisms integrated into the base model. This includes the baseline EfficientNet-B0 with no attention-mechanism, CBAM, and LSTL.

These mechanisms are designed to enhance the model's internal processing by enabling it to focus on the most relevant spatial and channel-wise features of retinal images.

The mediating variables explain how these attention mechanisms may influence performance. Specifically, Feature Representation Quality refers to how well the model captures fine-grained and subtle DR-related features, which is enhanced by attention-based modules. While Discriminative Power on Fine-Grained Classes describes the model's ability to differentiate between closely related stages of DR severity.

Finally, the dependent variables represent the measurable outcomes of model performance. These include accuracy, precision, recall, F1-Score, and FLOPs

Through this structure, the study aims to determine whether integrating the LSTL into the EfficientNet-B0 backbone yields statistically significant improvements over the baseline model and the CBAM-enhanced variant. The framework thus supports a

comparative approach, enabling quantitative analysis of classification performance and computational efficiency across the three model configurations.

Statement of the Problem

The objective of this study is to enhance EfficientNet-B0 by integrating a Local-Global Spatially-aware Transformer Layer (LSTL), which combines global and local attention mechanisms, to enhance the model's ability to detect subtle retinal features for improved Diabetic Retinopathy (DR) classification.

This aims to address the performance limitations of conventional CNN-based approaches when it comes to tasks that require fine-grained feature extraction. By embedding LSTL into EfficientNet-B0, the study seeks to improve the model's discriminative power with minimal computational overhead, making it more suitable for deployment in resource-constrained clinical settings.

In addition, to validate the effectiveness of LSTL as a viable alternative to other lightweight attention mechanisms, its performance will be compared against the Convolutional Block Attention Module (CBAM), another lightweight attention mechanism frequently used in medical image analysis.

The significance between these comparisons will be derived using a combination of Stratified K-Fold cross validation, the Friedman Test, and a Post-hoc (Nemenyi) Test.

Specifically, the study aims to answer the following questions:

1. What is the classification performance of the enhanced EfficientNet-B0 with LSTL compared to the unenhanced base EfficientNet-B0 model in terms of:
 - 1.1. Accuracy;
 - 1.2. Precision;
 - 1.3. Recall;

- 1.4. F1 Score; and
- 1.5. FLOPs?
2. What is the classification performance of the enhanced EfficientNet-B0 with LSTL compared to the enhanced EfficientNet-B0 model with CBAM in terms of:
 - 2.1. Accuracy;
 - 2.2. Precision;
 - 2.3. Recall;
 - 2.4. F1 Score; and
 - 2.5. FLOPs?
3. Is there a significant difference in classification performance between EfficientNet-B0 with LSTL when compared to either the baseline EfficientNet-B0 or EfficientNet-B0 with CBAM?

Hypothesis

(H0): There is no significant difference in classification performance between EfficientNet-B0 with LSTL when compared to either the baseline EfficientNet-B0 or the EfficientNet-B0 with CBAM.

Scope and Delimitations

This study will focus on developing an enhanced EfficientNet-B0 architecture for DR classification using retinal fundus images. Specifically, it aims to assess whether the integration of the LSTL can improve classification performance compared to two benchmarks: the baseline EfficientNet-B0 (without attention) and EfficientNet-B0 enhanced with CBAM.

To carry this out, the study utilizes the publicly available diabetic retinopathy (DR) dataset provided by APTOS. This dataset was selected to ensure standardized

benchmarking, support reproducibility, and enable consistent comparison across model configurations. It contains only retinal fundus images, and no supplementary patient information (e.g. age, sex, medical history, or clinical notes) will be included in the classification process.

The study is likewise limited to classifying images into five distinct DR severity levels, following the standard grading scale: No DR, Mild, Moderate, Severe, and Proliferative DR. It will not consider other types of eye diseases. And while broader generalizability to diverse datasets and clinical settings is recognized as an important direction for future research, it remains beyond the scope of this work.

Moreover, the study maintains a focused scope by restricting architectural modifications to attention mechanisms alone. It does not explore other forms of enhancement, such as alternative CNN backbones, ensemble methods, transformer-based architectures, or data-level strategies like preprocessing and sampling techniques. This decision ensures a controlled comparison between the LSTL and CBAM modules within the same base model.

In line with this architectural focus, the study also deliberately excludes post-hoc explainability methods such as saliency maps or Grad-CAM. While these techniques are valuable for understanding model behavior especially in clinical contexts, the primary objective of this research is to assess classification performance and computational efficiency, not to evaluate interpretability. Thus, explainability analysis is considered beyond the scope of this work.

In addition, only three specific model configurations are considered: the base EfficientNet-B0, EfficientNet-B0 with CBAM, and EfficientNet-B0 with LSTL. Other attention modules, particularly more advanced or computationally intensive transformer-

based variants, are excluded from this study, as the intention is to focus on lightweight solutions.

Furthermore, when it comes to measuring computational efficiency, the study relies on FLOPs as a metric for model complexity and computational requirements. While FLOPs provide a useful theoretical metric, the study does not capture other aspects of real-world deployment, such as inference speed on mobile hardware, memory consumption, or energy usage. These factors, though highly relevant for real-world implementation, are beyond the scope of this study.

Finally, it is important to note that while the models are trained and evaluated on a dataset labeled by medical professionals, the actual evaluation done in the study does not incorporate clinical validation or expert review from ophthalmologists. Consequently, while the results may potentially show promising improvements in classification accuracy, the model will not be tested in practical settings, and their clinical reliability remains to be established through future studies.

Significance of the Study

The outcomes of this study stand to offer meaningful contributions to the following sectors and communities:

Medical and Healthcare Industry. The result of the proposed study offers insights into the creation of lightweight models for classifying DR using retinal fundus images. By trying to enhance a lightweight model with attention mechanisms like LSTL, this study aims to ultimately contribute to the future development of more robust DR classification systems, especially in resource-limited areas where access to eye specialists is minimal. Additionally, the research also has the potential to improve early detection of DR, prevent vision loss, and promote

scalable telemedicine solutions by enabling fast and automated screening processes.

Computer Vision Communities. This study contributes to the ongoing exploration of lightweight attention mechanisms in deep learning. It introduces and evaluates the use of the LSTL within a compact CNN backbone, offering a novel direction for performance-efficient model design. By comparing it to the widely used CBAM module and baseline CNN, the study provides insights into the trade-offs between accuracy and computational cost in fine-grained medical classification tasks.

Public Health and Rural Communities. This study supports public health efforts by addressing a real-world challenge: the shortage of specialists in remote or underserved areas. By creating an accessible tool for DR detection that can potentially run on low-power devices, the research promotes health equity and the early prevention of blindness caused by diabetes and its complications that disproportionately affect populations with limited access to routine eye care.

Future Researchers in Medical Imaging. Future researchers in the field of automated diagnostics will also benefit from the methodology and findings of this study. The results can serve as a benchmark for integrating attention mechanisms into compact models and encourage further experimentation with model architectures that balance performance and efficiency. It may also guide future work on domain adaptation, transfer learning for different medical image types, or integrating clinical validation workflows into AI systems.

Definition of Terms

Diabetic Retinopathy (DR). Diabetic Retinopathy is the primary problem that the study aims to address. It is a chronic eye condition brought about by long-standing high blood glucose levels in Diabetes patients. It is identified by abnormal features that manifest in the retina such as cotton wool spots, neovascularization, hemorrhages, and microaneurysms. DR is divided into several stages, from mild, non-proliferative types to the more severe proliferative stages.

Microaneurysms. Microaneurysms These are the first visible indicators of diabetic retinopathy, showing up as tiny red dots on the retina due to the weakening of blood vessel walls. You'll usually find these little lesions near the macula, and they can be quite tricky to spot without high-resolution imaging techniques. Detecting microaneurysms is a vital part of diagnosing early-stage diabetic retinopathy, as their presence signals the onset of vascular damage in the retina.

Hemorrhages. Hemorrhages of diabetic retinopathy happen when the retinal blood vessels are broken, resulting in dark red or black spots developing in the layers of the retina. Hemorrhages are variable in shape and size based on severity, with larger hemorrhages denoting a more severe DR. Lesions exacerbate vision loss and, if left untreated, will cause abnormal blood vessel formation, aggravating the disease

Neovascularization. Neovascularization is abnormal and weak new blood vessel formation on the surface of the retina, usually in the proliferative phase of diabetic retinopathy. The new vessels are due to insufficient oxygenation of the retina but can leak and bleed easily, leading to destructive vision loss. Neovascularization complicates retinal detachment and needs to be treated promptly to avoid blindness.

Exudates. Lipid deposits that leak from damaged retinal blood vessels. They appear as bright, yellowish-white patches on the retina and are a common sign of diabetic retinopathy. "Hard exudates" will have well-defined borders, while "soft exudates" or "cotton-wool spots" are larger with indistinct borders and indicate areas of retinal ischemia.

Macular Edema. Is the swelling or thickening of the macula, the central part of the retina responsible for sharp, detailed vision. It occurs when fluid leaks from damaged blood vessels and accumulates in the macula. Macular edema is a common complication of diabetic retinopathy and a leading cause of vision loss in diabetic patients.

Fundus Photography. Fundus photography is a routine retinal imaging procedure that produces high-resolution images of the retina, optic disc, and blood vessels at the back of the eye. It is commonly employed for diabetic retinopathy screening and diagnosis through providing clear images indicating lesions like microaneurysms, hemorrhages, and exudates. These are the images that tend to be analyzed by automated systems using deep learning models for DR severity classification.

Local Attention. Is a specific component of attention mechanisms, which focus on capturing relationships within localized inputs. In the context of DR classification, local attention increases the chances of small and abnormal features like microaneurysms to be identified, as they are normally confined in specific areas of the retina.

Global Attention. Another component of attention mechanisms which will allow a model to establish relationships and dependencies across the entire input feature

map. By considering all features simultaneously, it enables the model to identify broad contextual information and long-range interdependencies, which can be crucial for understanding the overall severity and spread of Diabetic Retinopathy across the retina.

Chapter 2

REVIEW OF LITERATURE AND STUDIES

This chapter presents a review of literature that underpins the study's direction. It provides an in-depth overview of DR, outlines prior work on automated DR classification, identifies core challenges in existing deep learning-based approaches, and evaluates proposed enhancements from recent research. By examining these developments, the chapter establishes the basis for the study's proposed solution and its relevance in addressing performance and efficiency gaps in current models.

Diabetic Retinopathy Background

Diabetic Retinopathy (DR) is a severe complication of Diabetes Mellitus, affecting millions of people globally. In terms of effects, according to the Asia Pacific Tele-Ophthalmology Society (APTOS, 2019), DR is the leading cause of blindness among working-aged adults. Moreover, according to the World Health Organization (2023), it remains one of the primary leading causes of blindness worldwide. As per projections of Abdelmaksoud et al. (2021), Diabetes is expected to affect approximately 642 million adults worldwide by 2040, with DR affecting one of every three diabetics. Moreover, in another study, it is reported that people with DR might reach 191 million by 2030 (Congdon et al., 2012).

In the Philippines' case, DR continues to emerge as a critical and growing complication of diabetes, contributing significantly to the country's burden of vision-related disability. According to Cando et al. (2023), over 4.3 million Filipinos are diagnosed with diabetes, with an additional 2.8 million remaining undiagnosed, a gap that increases the likelihood of DR progression without timely intervention or treatment. The true magnitude of DR prevalence is therefore likely underestimated, especially among underserved populations

Furthermore, the Philippine Eye Research Institute (PERI, 2021) emphasizes that DR has become one of the leading causes of preventable blindness in the country. While treatment options are available, screening efforts remain limited due to challenges such as a shortage of trained specialists, lack of equipment, and uneven access to eye care services, especially in rural and remote areas. These limitations make it difficult to detect DR early, when treatment is most effective.

Early detection and classification of DR is critical for effective treatment and vision preservation as allowing it to progress to more severe stages can make it untreatable. However, one of the issues faced in this field is that manually diagnosing diabetic patients takes time for ophthalmologists. It is tedious, time-consuming, and prone to human error. Moreover, in rural and low-resource areas where medical screening is difficult to conduct and facilities are limited, this challenge is only further exemplified (APTOS, 2019).

According to Prasad et al. (2023), early screening can cut the risk of vision-threatening DR by more than half, because laser or anti-VEGF therapy is still effective while retinal damage is minimal. Bhaskaranand et al. (2016) states that manual grading not only varies from one specialist to another but also demands 5–10 minutes per image set, creating bottlenecks when thousands of patients need annual checks. And Cao et al. (2022) rural populations experience screening rates 25–40 percentage points lower than urban counterparts, owing to clinician shortages and long travel distances to eye-care centers.

Clinically, DR manifests as a range of retinal abnormalities, primarily observed through retinal imaging. Lesions commonly used to detect DR include: Microaneurysms (MA), Hemorrhages (HM), soft and hard exudates.

Microaneurysms is an early stage of diabetic retinopathy where small red round dots are present on the retina virtue of vessel wall weakness. Hemorrhages are diagnosed by

the presence of large spots on the retina with irregular margin sizes of upwards of 125 micrometers. Hard exudates are a consequence of plasma leakage and are visible as yellow spots on the retina caused by leakage of plasma. And soft exudates are a consequence of nerve swelling and are visibly white ovals on the retina.

Diabetic Retinopathy detection typically involves identifying 5 stages which are no DR, mild DR, moderate DR, severe DR, and proliferative DR (Mamtora et al., 2017; Murugesan et al., 2015):

1. No DR – No signs of retinal damage are visible.
2. Mild Non-Proliferative DR – Characterized by isolated microaneurysms.
3. Moderate Non-proliferative DR – Involves more microaneurysms, hemorrhages, and some vascular leakage.
4. Severe Non-Proliferative DR – Extensive vascular blockage occurs, accompanied by intraretinal microvascular abnormalities (IRMAs).
5. Proliferative DR (PDR) – Marked by neovascularization; fragile new vessels grow on the retinal surface, risking bleeding and retinal detachment (Shukla & Tripathy, 2023).

These stages may also be broadly categorized into two categories: Non-Proliferative DR and Proliferative DR. NPDR is marked by leakage of fluids and proteins due to retinal vessel damage, often accompanied by the previously mentioned lesions. And PDR, representing the advanced stage, is characterized by angiogenesis, where new blood vessels grow along retinal structures, particularly around vascular arcades, which poses serious risks to vision. According to Wong et al. (2018), accurate classification of DR severity levels helps identify high risk patients and hence helps mitigating possible problems.

Manual DR detection requires not only high-resolution imaging but also skilled interpretation. Even among experts, inter- and intra-grader variability can lead to inconsistent assessments. Furthermore, traditional screening workflows involve several stages, from optical coherence tomography (OCT) scans to fundus image analysis, demanding both time and resources. These limitations highlight the potential of automated, machine learning–based diagnostic tools to address issues of scalability, speed, and accuracy, especially in underserved areas where medical support is scarce.

Key Clinical Features for DR Detection

Diabetic Retinopathy (DR) is characterized by a variety of fine-grained and subtle retinal abnormalities that develop gradually and, if left undetected or untreated, may lead to serious vision loss. During an eye exam, several hallmark features may be observed, including microaneurysms, hemorrhages, hard exudates, cotton wool spots, intraretinal microvascular abnormalities (IRMA), and neovascularization (Shukla & Tripathy, 2023). These features are often small in size, low in contrast, and can be visually ambiguous, which presents challenges for both manual diagnosis and automated systems that rely on clear and high-level patterns.

Among the earliest and most telling signs of DR are microaneurysms (MAs) which are tiny swellings in the walls of retinal blood vessels. Typically ranging from 10 to 125 microns, they manifest as small, round red spots in fundus images and are often found near the macula, where their presence can threaten central vision (Manjaramkar & Kokare, 2017). Because of their size and similarity to dot hemorrhages, MAs are difficult to detect without advanced imaging tools like fluorescein angiography, where they appear as bright, pinpoint leaks. Early identification of MAs is important but is challenging, especially for models like CNNs that may overlook these subtle, pixel-level variations due to their architecture.

Beyond MAs, hemorrhages and exudates are also key indicators. Hemorrhages appear as dark red or black blotches caused by bleeding from damaged vessels and can vary significantly in size and shape, making them harder to classify consistently. In contrast, exudates are bright yellow or white spots with well-defined edges, often forming near leaky capillaries. While exudates are somewhat more distinct, their appearance still depends heavily on the surrounding retinal context, which can obscure their presence, especially in lower-quality or low-contrast images (Chandhakanond & Aimmanee, 2025).

Another frequently overlooked indicator is the cotton wool spot (CWS) which is a soft, white lesion caused by localized retinal ischemia. These lesions are incredibly small, with average sizes around 0.33 mm^2 , and often blend into the retinal background, making them particularly elusive in standard fundus photography (Mahdjoubi et al., 2020). Similarly, intraretinal microvascular abnormalities (IRMA), which are fine, abnormally twisted vessels, are often only 31 micrometers wide and form deep within the retinal layers. Their visibility in color fundus images is limited, requiring sophisticated imaging techniques such as swept-source OCT angiography for reliable detection (Ding et al., 2024). These extremely delicate features, due to their minute scale and diffuse appearance, challenge traditional CNNs, which are optimized for detecting coarse spatial hierarchies rather than intricate vascular details.

The detection difficulties associated with DR share parallels with those encountered in other medical domains. For instance, in oncology, the accurate localization of subtle lesions plays a pivotal role in cancer diagnosis, staging, and prognosis (Zhang et al., 2024). Yet, these features are often masked by heterogeneous backgrounds and complex tissue textures, leading to high variability in detection performance (Russo et al., 2024).

DR grading relies on similarly precise detection of minute features—requiring clinical tools such as biomicroscopy, fundus imaging, OCT, and Doppler OCT (Das et al., 2022). However, these techniques are time-intensive and subject to human error, particularly due to inter-observer variability in interpreting fine features like MAs and neovascularization (Krause et al., 2018). The granular nature of these lesions and their importance in determining disease progression highlight a core limitation of conventional CNN-based approaches: their difficulty in capturing global context while simultaneously localizing small-scale anomalies.

These limitations underscore the critical need for more sophisticated and precise automated systems, especially those that are capable of learning both local detail and broader spatial context. Addressing this challenge is essential for improving early diagnosis and ensuring timely clinical interventions.

Traditional Methods Used in DR Classification

Diabetic Retinopathy grading has conventionally depended on manual and semi-automatic methods that need professional interpretation. These have been extensively applied in clinical practice prior to the advent of deep learning-based methods. Although effective, they tend to be associated with issues of time requirement, subjectivity, and accessibility.

One of the most common conventional approaches is fundus photography, which produces high-quality images of the retina. With this method, ophthalmologists can visually examine retinal pathologies like microaneurysms, hemorrhages, and exudates. Fundus photography has been a standard for DR diagnosis over the past several decades, giving useful information regarding disease progression (Alyoubi et al., 2021). Nonetheless, manual grading of fundus images can be time-consuming and subject to variation between graders (Krause et al., 2018).

Another widely employed method is Optical Coherence Tomography (OCT), a non-invasive imaging technique that offers cross-sectional images of the retina, assisting in the detection of changes in retinal thickness and fluid leakage. OCT has been crucial in detecting early manifestations of DR, especially when subtle retinal alterations are hard to identify using fundus photography alone (Kapoor & Arora, 2021). Although effective, OCT is costly and not always available in resource-poor environments.

Fluorescein Angiography is another conventional technique employed for DR grading. This method involves the injection of a fluorescent dye into the blood to make abnormalities in blood vessels in the retina stand out. It is especially helpful in identifying neovascularization and the severity of DR (Das et al., 2022). The procedure is invasive and can be painful for patients, making it less suitable for widespread application. Adaptive Optics Imaging has also been explored as a method for enhancing retinal imaging resolution, allowing for detailed visualization of microvascular changes. Though its high cost and technical complexity have limited its adoption in routine clinical practice, this technique improves the ability to detect early signs of DR (Chandhakanond & Aimmanee, 2025).

Machine Learning Methods for DR Classification

Traditional machine learning (ML) models such as Support Vector Machines (SVM), k-Nearest Neighbors (k-NN), Decision Trees, and Random Forests have played a significant role in classification tasks across various domains, including early-stage medical imaging. Despite their widespread use, these methods exhibit notable shortcomings when applied to complex image-based tasks like Diabetic Retinopathy (DR) classification.

One significant drawback of traditional ML models is their reliance on manual feature extraction, which requires extensive domain expertise and preprocessing efforts.

DR images vary significantly in terms of lesion size, texture, illumination, and resolution. Extracting relevant features, such as microaneurysms, hemorrhages, or exudates, using manual or rule-based techniques is not only labor-intensive but also requires extensive preprocessing and expert tuning, making it prone to subjective bias and inconsistencies (Khojasteh et al., 2018). Consequently, these handcrafted methods struggle to generalize across diverse datasets, leading to suboptimal classification accuracy, particularly in real-world applications. In contrast, deep learning models, particularly Convolutional Neural Networks (CNNs), overcome these limitations by learning these features directly from raw pixel data, reducing the need for manual intervention (Gulshan et al., 2016). Their study demonstrates that CNN-based models can automatically extract predictive features from retinal fundus images without predefined lesion-based characteristics. By leveraging hierarchical feature learning, CNNs progressively capture intricate patterns, from basic edge structures to complex retinal abnormalities. This ability of CNNs significantly improves sensitivity and specificity in diabetic retinopathy classification, offering a more efficient method compared to traditional learning methods.

Traditional ML models struggle with high-dimensional data like images, which often contain tens or hundreds or thousands of features (pixels). To manage this complexity, traditional ML models such as SVM and Logistic Regression often rely on dimensionality reduction techniques (Yang et al., 2025). Additionally, Logistic Regression requires feature selection techniques (Wan et al., 2025), both of which can inadvertently remove informative patterns relevant to DR classification. This limitation reduces the model's ability to detect fine-grained variations necessary for distinguishing between stages of DR.

Furthermore, another challenge lies in their susceptibility to domain shifts and intra-class variability. Factors such as differences in patient demographics, imaging conditions, and diagnostic criteria introduce inconsistencies that impact model

performance. Traditional algorithms frequently suffer from performance degradation on unseen domains, making robust feature-learning strategies necessary—tasks that deep learning models inherently handle more effectively (Xia et al., 2024).

In addition, most traditional ML models rely on manually designed feature extractions and separate optimization stages. Each stage (e.g., preprocessing, feature selection, classification) is developed and optimized separately, which can introduce cumulative error and inefficiencies. Deep learning models employ end-to-end training to automatically extract hierarchical patterns from retinal images. This approach enhances feature representation and improves classification accuracy, making deep learning models more effective for diabetic retinopathy (Chetoui & Akhloufi, 2020).

Finally, the scalability of traditional ML models is limited, especially when handling large datasets such as EyePACS and Messidor. Algorithms like k-NN and SVM often face prohibitive training time and memory requirements. In contrast, CNN architectures optimize computational efficiency through parallelization on GPUs, enabling faster processing and improved classification accuracy for diabetic retinopathy detection (Lakshmi et al., 2024).

In summary, while traditional machine learning methods laid the foundation for early image classification efforts, their structural limitations make them less suitable for nuanced, data-intensive tasks such as diabetic retinopathy classification. Their reliance on manual feature extraction, dimensionality reduction, and separate optimization stages limits efficiency and scalability. Deep learning, particularly CNNs, have largely superseded them due to their ability to automatically learn hierarchical features from raw images, handle high-dimensional data, optimize classification through end-to-end training, and improve accuracy and robustness.

General Overview of Deep Learning in DR Classification

In recent years, advances in deep learning (DL) have positioned it at the forefront of computer vision applications in medical analysis. And while no DL-based system has yet been widely adopted for clinical diagnosis, numerous studies in medical image analysis have demonstrated promising results, showing the significant progress in diagnostic and clinical decision support. In ophthalmology, and particularly in the classification of diabetic retinopathy (DR), where the detection of subtle features is crucial for an accurate diagnosis, DL-based methods have long outperformed traditional machine learning (ML) techniques and has become the leading paradigm due to their performance (Litjens et al., 2017).

One of the key strengths of deep learning in medical image analysis is its ability to automatically learn hierarchical feature representations from raw image data, making it particularly well-suited for learning and detecting the features. As discussed in earlier sections, effective detection of diabetic retinopathy (DR) depends on identifying various small, localized lesions such as microaneurysms, hemorrhages, and exudates. These lesions are challenging to detect due to their variability in shape, size, color, and spatial location.

Traditional machine learning (ML) methods, such as support vector machines and random forests, often struggle in this context. As they rely on handcrafted features, which can limit their ability to generalize across different patient populations, imaging conditions, and camera types (Gulshan et al., 2016; Rajalakshmi et al., 2018).

In contrast, deep learning—particularly through convolutional neural networks (CNNs)—eliminates the need for manual feature extraction. CNNs build increasingly abstract representations layer by layer, learning directly from pixel-level data. This enables them to adapt to visual cues and complex hierarchical patterns found in retinal fundus

images. Such capability is important in DR classification, where key indicators like microaneurysms and hemorrhages may be small, spatially dispersed, or affected by variations in lighting and contrast. By progressively learning low-level features such as edges and blobs, and eventually high-level semantic representations, CNNs are uniquely equipped to capture the fine-grained patterns essential for accurate diagnosis (LeCun et al., 2015).

Another key advantage of CNNs lies in their spatial inductive bias. As through convolutional filters, CNNs maintain the spatial structure of retinal fundus images, which is an essential aspect in the diagnosis of diabetic retinopathy (DR). The spatial distribution, clustering, and relative positioning of lesions in relation to anatomical landmarks such as the optic disc or macula play a significant role in determining DR severity. Because of their local receptive fields and hierarchical architecture, CNNs are capable of capturing both local features and broader global context. This is a considerable strength over traditional ML models, which typically require extensive pre-processing or explicit segmentation pipelines to incorporate spatial information (Gulshan et al., 2016; Pratt et al., 2016).

Beyond spatial awareness, convolutional neural networks (CNNs) have also demonstrated greater resilience to the noise and variability commonly found in real-world medical imaging data. Retinal fundus images captured across different clinics often suffer from issues such as poor lighting, motion blur, inconsistent fields of view, and imaging artifacts like dust or reflections. Traditional models tend to perform poorly under such conditions unless trained on highly curated datasets. In contrast, CNNs have shown a strong capacity to generalize when trained on large and diverse datasets. Their ability to learn invariant and discriminative features enables them to maintain robust performance despite these variations (Shen et al., 2017).

Lastly, CNNs offer a high degree of extensibility. Modern architectures support the integration of modules such as attention mechanisms, residual connections, and even hybrid components that combine convolutional layers with transformer-based designs. These modular enhancements can significantly improve performance without requiring the entire system to be redesigned. In the context of DR classification, models have evolved from early CNN-based approaches to more sophisticated variants such as EfficientNet, DenseNet, and hybrid CNN-transformer architectures. Each of these aims to optimize the trade-off between classification accuracy, computational efficiency, and inference speed (Tan & Le, 2019; Li et al., 2021).

Given the complexity of DR as a progressive, feature-rich disease with substantial inter-patient and intra-class variability, convolutional neural networks offer a sound method for automated classification systems, as their capacity to generalize across data sources, capture subtle features, and scale to large datasets positions them as one of the most appropriate tools for the task.

Existing CNN-Based Models for DR Classification

The application of convolutional neural networks (CNNs) in tasks that need fine-grained feature extraction like diabetic retinopathy (DR) classification has shown promise; however, current models continue to fall short of the high accuracy thresholds that models of medical nature need for clinical deployment. According to Kim (2021), achieving an accuracy that is less than 90% is a limitation. As in medical diagnostics, particularly in fields like ophthalmology where early intervention can prevent irreversible vision loss, a DR classifier must be accurate, given the consequences of misclassification.

While it is crucial in medical diagnostics to obtain high accuracies to be able to ensure patient safety, several studies have still reported lower accuracies, which indicates the need for improved models.

In a study conducted by Kim (2021) aimed at identifying the most effective pre-trained CNN model for diabetic retinopathy (DR) classification, five well-known architectures were evaluated: MobileNet, VGG16, VGG19, Inception V3, and Inception ResNet V2. The baseline CNN model used in the study achieved only 10% accuracy, which highlights the limitations of training CNNs from scratch for this task. To address this, the researchers turned to transfer learning with models pre-trained on ImageNet. Among the tested architectures, MobileNet achieved an accuracy of 66.04%, VGG16 reached 71.21%, while its improved version, VGG19, slightly outperformed it with 73.35%. Inception V3 performed better still, achieving 76.84%, and InceptionResNetV2—which they modified with GRU—yielded the highest accuracy at 83.57%. However, as we've mentioned, this still isn't enough for tasks like DR classification where accuracy is paramount.

In another study by Ganesh et al. (2021), they also explored the feasibility of transfer learning with three architectures: namely Xception, InceptionResNetV2, and EfficientNetB3, which were all pre-trained on ImageNet for the task of DR diagnosis. For the 5-ary classification, the Xception model was able to reach the highest accuracy of 69%, the InceptionResNetV2 was able to achieve an accuracy of 64%, and the EfficientNetB3 obtained an accuracy of 62%. However, while the EfficientNetB3 performed the worst of the three, the researchers stressed that EfficientNetB3 had only 12 million parameters when compared to the 22 million and 55.8 million parameters of Xception and InceptionResNetV2. And so, despite being smaller, it achieved comparable results with faster inferences, training, and convergence rates making it a viable solution for automated diagnostic systems.

The study concluded by emphasizing the role of early-stage DR detection in improving patient recovery outcomes. Given the growing need for efficient diagnostic

systems, the researchers highlighted that state-of-the-art deep learning techniques are being increasingly adopted due to their potential in automating and accelerating medical diagnoses. Notably, they stressed that the time required to train, validate, and test these models significantly affects research progress in the field. As such, EfficientNet, with its lightweight architecture and competitive accuracy, was recommended as a strong alternative to traditional deep CNNs, as its ability to converge faster, reduce computational load, and produce reliable diagnostic outcomes supports its viability in real-world applications where speed and resource efficiency are important (Ganesh et al., 2021).

Haripriya et al. (2022) also proposed a CNN-based approach for diabetic retinopathy classification by modifying the InceptionV3 architecture. Their method involved adjusting the batch size and optimizing the model for better performance on a dataset of 3,724 retinal images, which they then classified into five DR severity levels. The modified model achieved an accuracy of 80%, outperforming other baseline architectures such as VGG16, ResNet50, and DenseNet121 tested in the same study. This further demonstrates the potential of customized transfer learning models in enhancing diagnostic accuracy for diabetic retinopathy detection. However, again, this falls short of our desired 90% accuracy threshold.

While prior studies have largely focused on comparing the performance of multiple pre-trained architectures for DR classification, other research has taken a more focused approach, aiming instead to enhance the performance of specific models by addressing specific limitations found in DR classification. One such limitation is the issue of class imbalance, which is common in medical imaging datasets. This imbalance, where certain severity levels are underrepresented, can skew the model's learning process, which results in reduced sensitivity to minority classes and ultimately lower classification accuracy.

To address this issue, a recent study titled "Diabetic Retinopathy Detection Using VGG-16 Deep Learning Architecture" by Yatam & Gera (2024) proposed an improvement to the standard VGG-16 model by incorporating a data balancing technique known as Pre-SMOTE (Pre-Synthetic Minority Over-sampling Technique). Unlike other studies that explored various architectures, this study narrowed its focus to enhancing one architecture, VGG16, by improving the quality and distribution of the training data. By applying pre-SMOTE to generate synthetic samples for underrepresented classes before model training, the researchers were able to transform an imbalanced dataset into a more uniform one, which in turn enabled more robust learning.

The results were notable: the VGG-16 model, when trained without any balancing technique, achieved an accuracy of 71.65%. However, after applying Pre-SMOTE, the model's accuracy rose significantly to 79.99%, demonstrating the role that data distribution plays in model performance. This improvement highlights that data imbalance is a substantial bottleneck in DR classification performance. Nevertheless, even with these enhancements, the model still fell short of the 90% accuracy threshold often regarded as desirable for clinical diagnostic tools, suggesting that while techniques like Pre-SMOTE can provide meaningful gains, further improvements in architecture, data preprocessing, or attention mechanisms may still be necessary to meet real-world reliability standards.

In a more recent effort to develop a comprehensive diagnostic pipeline, Long et al. (2024) in their study, proposed an end-to-end deep learning system based on EfficientNetB0 for the classification of diabetic retinopathy and detection of diabetic macular edema (DME). The study introduced a two-stage framework: the first stage performed a five-class classification task to distinguish between normal retinas, varying DR severity levels, and post-laser treatment images; the second stage further analyzed

abnormal cases to detect the presence of DME. The model achieved 70.36% accuracy (AUC: 0.9062) in the first stage and 72.58% accuracy (AUC: 0.7530) in the second.

While the study demonstrated strong discriminative power, especially in terms of AUC, the moderate accuracy levels highlight a performance ceiling that limits its clinical readiness. Despite the large dataset of over 19,000 images and the use of Grad-CAM for interpretability, the relatively low classification accuracy suggests that there is still considerable room for improvement—particularly for real-world deployment where higher precision and reliability are crucial. Similar to the previous study, this indicates the need for further exploration of architectural improvements that can bridge the gap between diagnostic accuracy and clinical applicability.

In another recent study, Lin and Wu (2023) also aimed to enhance the performance of the ResNet-50 model for diabetic retinopathy classification. Likewise, their research focused on improving the effectiveness of a single model by addressing two key challenges: inconsistencies in image quality and the tendency of CNNs to overfit on small or imbalanced medical datasets.

They explored preprocessing techniques and introduced their structured Standard Operating Procedure (SOP) for preprocessing the fundus images. This included removing the black borders typically found in raw fundus images, applying circular cropping to focus only on the retinal region, and using histogram equalization to enhance image contrast.

Additionally, to combat overfitting, they proposed a revised ResNet50 architecture. Regularization was applied, specifically, they applied both L1 and L2 regularization to help constrain the learning process and reduce model complexity. These were implemented using the `kernel_regularizer`, which penalizes the weights of the layer's kernel, and

activity_regularizer, which applies a penalty to the layer's output. This dual regularization approach was designed to enhance the model's ability to generalize to unseen data.

The effectiveness of these revisions was measured by comparing the original and revised versions of ResNet-50. The original model achieved a training accuracy of 89.81% and a test accuracy of 75.67%. In contrast, the revised version achieved a slightly lower training accuracy of 83.95%, indicating better generalization, but a comparable test accuracy of 74.32%. This result suggests a reduced overfitting tendency and demonstrates the value of architectural tuning and preprocessing in enhancing DR classification performance. However, the overall classification performance still remains below the desired threshold for clinical deployment. Again, this reinforces the need to explore further improvements.

In another effort to enhance DR classification, Gangwar and Vadlamani (2020) proposed a hybrid deep learning model that combines transfer learning with custom convolutional layers. Their approach involved leveraging the strengths of Inception-ResNet-v2 as a feature extractor while appending additional CNN layers tailored for DR detection. This modification aimed to refine and deepen the representational capacity of the network.

Their hybrid model was evaluated on two widely used datasets: Messidor-1 and APTOS 2019. There, it achieved a test accuracy of 72.33% on Messidor-1 and a notably higher 82.18% on the APTOS dataset, outperforming several previously published models on the same benchmarks at the time. Although their study delivered noticeable improvements, its performance remains below optimal standards for clinical reliability. Similar to before, this highlights the ongoing need to explore additional enhancements, whether through architectural innovations, refined preprocessing techniques, or more advanced learning strategies.

A recent study done by Aasrith et al. (2024) also took a focused approach by leveraging the EfficientNet-B0 architecture for the task of diabetic retinopathy classification. Recognizing the importance of balancing performance with model efficiency especially for deployment on resource-constrained devices, they employed an optimized pipeline using a preprocessed Kaggle dataset of over 2,200 high-resolution fundus images categorized into five DR severity levels. Their version of EfficientNet-B0 achieved a peak validation accuracy of 84%, which is notably higher than many previously reported models. In addition to its relatively high accuracy, the model demonstrated strong generalization capacity and efficient convergence during training. This reinforces the idea that lightweight architectures like EfficientNet-B0, when properly tuned, can offer a compelling balance between diagnostic performance and deployment practicality, particularly for real-time medical diagnostic systems.

On a slightly different note, Huang et al. (2023) also conducted a systematic investigation into the components of the ResNet-50 architecture that most significantly influence diabetic retinopathy (DR) grading performance and offered insights that are methodological rather than architectural. Rather than proposing a novel model, their work focuses on dissecting which preprocessing steps and training strategies most affect classification accuracy. For example, they evaluated two preprocessing techniques common in DR classification: Graham's method, which enhances image contrast by subtracting the local average color, and CLAHE (Contrast Limited Adaptive Histogram Equalization), which enhances local contrast. However, for some reason, both approaches hurt the performance of their model.

Additionally, the study tackled class imbalance using three sampling strategies: instance-balanced sampling, class-balanced sampling, and progressively balanced sampling. The progressively balanced approach, which transitions from class-balanced to

instance-balanced over time, was found to best mitigate bias while maintaining model generalizability. Overall, their findings underscore that optimizing training settings can also yield significant performance gains even without modifying the network architecture. As they emphasized, their findings are not specific to ResNet-50 and can be generalized to other CNN-based architectures used in DR detection and grading tasks.

Across the many efforts reviewed, it is clear that diabetic retinopathy classification using CNNs has progressed steadily. However, they still consistently fall short of the reliability demanded for clinical deployment. Despite innovations in model architecture, preprocessing, and data balancing, most solutions still hover below the ideal performance threshold, especially in terms of classification accuracy. This suggests that future work can be directed on not only refining existing strategies but also exploring different directions that can bridge this performance gap.

CNN Limitations for Small Feature Extraction

Convolutional Neural Networks (CNNs), while widely adopted and highly effective for general image classification, have been shown to face limitations when applied to fine-grained tasks such as diabetic retinopathy (DR) detection. These models are designed to learn hierarchical representations, where deeper layers capture more abstract, global features. However, this global emphasis can come at the cost of overlooking small, localized details that are important in medical imaging, particularly in cases where subtle retinal lesions, such as microaneurysms and hemorrhages, signal early-stage DR (Kalra et al., 2021).

EfficientNetB0, a widely used lightweight CNN, inherits these limitations due to its architecture that reduces spatial dimensions to maintain computational efficiency. As, while this improves speed and parameter efficiency, it may inadvertently discard the fine details necessary for distinguishing between mild and moderate DR stages.

These challenges stem from the way CNNs inherently handle feature extraction. As convolutional filters and pooling layers emphasize dominant textures and shapes while discarding finer details, especially in deeper layers, this makes them less suited for tasks where class-discriminative features are sparsely localized (Kalra et al., 2021). Additionally, as also mentioned in another study, it is said that because of these same pooling operations and convolutional filters that prioritize dominant patterns, CNNs often struggle with subtle intra-class variations due to poor sensitivity to small-scale features (Reilly, 2024).

Furthermore, CNNs do not inherently model long-range dependencies, which limits their contextual awareness and hampers their performance on complex visual scenes (Zhao et al., 2023).

In a broader context, the limitations of CNNs in texture-based or fine-grained classification tasks have also been observed across various domains. In this, Kalra et al. (2021) emphasized that traditional CNNs may underperform when local patterns carry crucial class-defining information, such as in dermatological or retinal image analysis. They mentioned that this problem arises because early CNN layers may lack sufficient receptive fields to contextualize small features, while later layers may generalize them away due to coarse abstraction. Moreover, such challenges have also been observed in other domains like dermatological and histopathological image analysis, where local visual patterns are key to accurate diagnosis.

Attention Mechanisms in Computer Vision

To address the issues mentioned above that are inherent in CNNs, one such method that have been explored is the integration of attention mechanisms into CNN-based models. Attention mechanisms are introduced as a way to enhance feature discrimination by allowing models to focus on the most relevant parts of an image.

Unlike fixed receptive fields in CNNs, attention mechanisms adaptively weight spatial and channel-wise features based on their importance, enabling the network to retain even small but significant details (Zhao et al., 2023).

They enable the network to dynamically reweight feature maps and emphasize critical areas while suppressing irrelevant ones. Using them allows the model to improve its discriminative power for small features. In the context of medical imaging specifically, attention-mechanisms are beneficial, as it allows models to focus on lesion-prone areas of retinal scans without being overwhelmed by irrelevant background textures. In fact, attention-augmented CNNs have been shown to improve diagnostic accuracy by focusing on lesion-prone regions of the retina (Zhao et al., 2023).

In their study, Zhao et al. (2023) demonstrated that adding self-attention and hybrid spatial-channel attention modules significantly improved a CNN's ability to locate discriminative regions and extract meaningful features, thereby enhancing classification performance in medical image datasets. Similarly, in another study done by Reilly (2024), they reported that adding self-attention to their CNN model improved its sensitivity to features, which suggests that attention mechanisms can directly mitigate the spatial information loss commonly associated with downsampling in CNNs.

However, despite the advantages of self-attention mechanisms in enhancing feature discrimination, they come at the cost of significant computational overhead. The standard self-attention mechanism computes pairwise interactions between all input tokens, resulting in a time and space complexity that scales quadratically with the input length ($O(n^2)$) (Hegde et al., 2023). This quadratic complexity becomes particularly burdensome in high-resolution medical imaging tasks, where the number of tokens (e.g., image patches) is substantial. Such computational demands can hinder the deployment

of attention-augmented models in resource-constrained environments, such as in real-time clinical settings.

To mitigate these challenges, other researchers have also proposed various efficient attention mechanisms. For instance, the Linformer approximates the self-attention matrix with a low-rank matrix, reducing the complexity from $O(n^2)$ to $O(n)$ (Wang et al., 2020). Similarly, the Performer employs kernel-based methods to achieve linear complexity in self-attention computations (Choromanski et al., 2021). These innovations aim to retain the benefits of attention mechanisms while making them more computationally feasible for practical applications.

However, both Linformer and Performer come with their own trade-offs. While Linformer's approach reduces complexity, it may sacrifice the model's ability to fully capture long-range dependencies, especially in dense feature spaces. Additionally, a study has noted that Linformer can underperform on tasks requiring high fidelity attention, like fine-grained vision tasks (Tay et al., 2020). It was also mentioned that the performance of Linformer is highly dependent on the choice of projection dimension (k). So to say, a dimension that's too small might severely degrade performance, while too large defeats the purpose of the linear approximation (Wang et al., 2020).

On the other hand, Performer uses kernel-based random feature mappings to approximate the softmax attention. This stochastic approximation introduces noise, especially in smaller models or datasets, which can hurt performance if not well-tuned. Additionally, the random feature approach can make training unstable without careful initialization and scaling. Moreover, it also requires non-trivial implementation compared to standard attention, which may limit its adoption or reproducibility in certain domains (Choromanski et al., 2021).

Beyond these, there have also been other attention mechanisms that have been developed. Zheng et al. (2023) mentions that current attention methods can be divided into two kinds: channel attention and spatial attention methods. Specifically, the channel attention method emphasizes effects of channel features on the whole CNN. While the spatial attention method treats pixels of all dimensions at the same location as a whole and its weight is learned via each pixel at each location.

The Convolutional Block Attention Module (CBAM) for one, which was introduced by Woo et al. (2018), is a lightweight and modular attention mechanism designed to enhance convolutional neural networks (CNNs) by refining feature representations through both channel and spatial attention. Applied sequentially, these two stages allow the network to first emphasize what features are most informative and then focus on where in the spatial domain these features reside.

However, despite its advantages, CBAM has not seen widespread adoption compared to other attention mechanisms, with one reason being its relative age. Developed in 2018, CBAM has since been overshadowed by newer modules such as Squeeze-and-Excitation (SE), Bottleneck Attention Module (BAM), and Transformer-based self-attention mechanisms. Additionally, CBAM's performance is somewhat domain-dependent; as while it excels in some visual tasks, it can be outperformed by SE or non-local blocks in others, particularly where joint or global attention modeling proves more effective. Finally, its sequential attention design (applying channel and spatial attention one after the other) may limit its flexibility compared to architectures that compute these attentions jointly or incorporate long-range dependencies, as seen in Transformer-based models.

Another widely used lightweight attention module is the Squeeze-and-Excitation (SE) block, which was introduced to recalibrate channel-wise feature responses by

explicitly modeling interdependencies between channels (Hu et al., 2018). The SE module achieves this by using a global average pooling operation to "squeeze" spatial information, followed by a pair of fully connected layers that "excite" the channel descriptors and generate reweighting coefficients. These coefficients are then used to modulate the original feature maps, enhancing the representation of informative channels while suppressing less relevant ones. SE is particularly efficient in terms of both parameters and computation, making it an appealing enhancement to CNNs, especially in resource-constrained scenarios. Its effectiveness has led to widespread adoption and integration into numerous model families. However, SE exclusively focuses on channel attention and does not account for spatial correlations between features, potentially limiting its ability to localize small discriminative regions—an important aspect in medical imaging tasks like diabetic retinopathy classification.

On the other hand, the Bottleneck Attention Module (BAM) offers a more balanced approach by incorporating both spatial and channel attention in parallel. Proposed as an extension to residual networks, BAM inserts attention modules into the bottleneck blocks of CNNs, enabling the network to refine features at multiple depths (Park et al., 2018). Its parallel structure allows it to jointly capture spatial saliency and inter-channel dependencies, making it more expressive than SE while being more computationally manageable than full self-attention layers. BAM's design is particularly useful in scenarios where both local and global contextual information are important. However, similar to CBAM, BAM's task-specific performance may vary, and its integration often requires tuning to match the architecture and input resolution, which can limit its plug-and-play versatility.

While each of these attention modules brings distinct advantages, they tend to emphasize either spatial or channel features or rely on sequential designs that may miss

richer interactions between the two. Moreover, most lightweight modules are either too narrow in focus (like SE) or lack the ability to jointly model local and global dependencies effectively.

Zhou et al., (2023) highlighted that despite the advancements brought by attention mechanisms in enhancing feature discrimination within lightweight convolutional neural networks (CNNs), challenges still remain. They mentioned that while lightweight CNNs are favored in medical image analysis for their efficiency and reduced risk of overfitting, their feature extraction capabilities are often inferior to their heavyweight counterparts. And although attention mechanisms show promise, they noted that existing attention modules, such as the Squeeze-and-Excitation (SE) and Convolutional Block Attention Module (CBAM), still possess insufficient non-linearity, which affects the ability of CNNs to capture key features in medical images.

This observation underscores a significant gap in current methodologies, as there remains a pressing need for attention mechanisms that not only enhance feature discrimination but also introduce sufficient non-linearity to improve the representational capacity of lightweight CNNs in medical image analysis.

It presents an opportunity in tasks like diabetic retinopathy classification, where fine-grained patterns and broader contextual cues are both important, but at the same time, we have to achieve those without incurring too much computational costs. And so, in this study, we aim to explore the integration of a novel attention module in a CNN backbone like EfficientNetB0, in order to improve their performance in detecting and classifying subtle pathological features in medical images. To present an improvement over existing methods and develop one that is able to capture both granular detail and broader structure, while remaining computationally practical for real-world deployment.

Transformers in DR Classification

Over the past few years, CNNs have been at the forefront of computer vision applications in the detection of various retinal diseases (RD). However, despite their relative successes, the emergence of Vision Transformers (ViT) has shifted efforts in RD model development from CNNs to ViTs. ViT-based models differ from CNNs in their approach to processing images. As, instead of scanning images using local receptive fields as CNNs do, ViTs divide images into fixed-size patches and apply self-attention mechanisms across these patches to model long-range dependencies. This architectural design has allowed ViTs to outperform traditional CNNs in several retinal disease classification tasks by capturing more global and contextual information.

However, this performance often comes at the cost of increased computational and data requirements. As unlike CNNs, ViTs are less inductive in nature, meaning that they do not inherently assume locality or translation invariance, and therefore require large amounts of data and extensive training to generalize effectively. More importantly, they demand significantly more memory and processing power during both training and inference. This makes them less practical in clinical environments, especially in remote or resource-limited settings where hardware constraints is a critical concern.

In the study “Comparative Analysis of Vision Transformers and Conventional Convolutional Neural Networks in Detecting Referable Diabetic Retinopathy” by Hui et al. (2024), the authors conducted a detailed comparison between Vision Transformer (ViT) models and traditional Convolutional Neural Networks (CNNs) for diabetic retinopathy (DR) classification. Their results showed that ViT models outperformed CNNs in key performance metrics such as area under the curve (AUC) and sensitivity. However, the authors also acknowledged a significant limitation: the high computational demands of ViTs restricted their ability to explore the optimal scaling and tuning of each architecture.

Moreover, they noted that due to the relative novelty of ViTs, further comprehensive evaluation, including parameter optimization and fine-tuning using real-world clinical data, is essential before ViTs can be reliably compared to well-established CNN architectures in practice.

Similarly, in the study “ViT-DR: Vision Transformers in Diabetic Retinopathy Grading Using Fundus Images” by Mohan et al. (2022), the researchers sought to create a ViT-based model for DR classification, which then achieved an accuracy of 87.41%, surpassing the performance of several CNN-based approaches referenced in our previous related literature. Nonetheless, the marginal improvement in classification accuracy came at the expense of significantly greater computational requirements.

As Ganesh et al. (2021) emphasized, DR classification is a task where computational efficiency matters, particularly if the goal is to integrate diagnostic tools into clinics that lack access to high-end computing infrastructure. Similarly, Aasrith et al. (2024) stressed the importance of designing models that can operate on resource-constrained hardware. In their study, they highlighted that while more complex models like EfficientNet-B7 may achieve better accuracy compared to models like EfficientNet-B0, they come with higher costs in terms of power consumption, which raises issues about deployment feasibility and long-term operational expenses. And so, their findings highlight the need for lightweight, cost-effective models that strike a balance between performance and deployability.

These concerns have also been further echoed in other literature. Islam et al. (2023) note that high computational demands can act as a barrier to the real-world adoption of DR classification models, particularly in low-resource healthcare systems. Additionally, Rezaee and Farnami (2025) also further emphasized that future research should prioritize optimizing models for deployment in constrained environments. As in this

context, the goal of automated DR screening systems must extend beyond achieving high accuracy; it must also include considerations of cost-effectiveness and accessibility.

Overall, while ViTs may offer modest performance gains over CNNs in DR classification, their substantial computational overhead remains a barrier to clinical adoption. This again highlights the importance of balancing model performance with efficiency and the need for exploration of alternative attention mechanisms which are more efficient, particularly for applications aimed for deployment in low-resource environments.

Architecture of EfficientNet

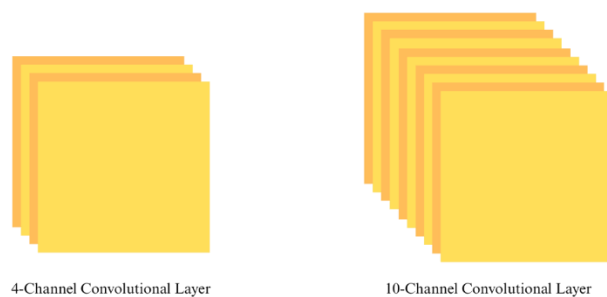
EfficientNet (Efficient Network) is a family of CNN models developed and introduced by Mingxing Tan and Quoc V. Le in 2019 under Google AI. EfficientNet, as the name suggests, is an efficient Neural Network capable of performing on par, or sometimes even better at some tasks compared to other lightweight and scalable Convolutional Networks such as MobileNets and Resnets.

EfficientNet builds upon two key-concepts for accuracy improvement: model scaling and image scaling (Tan and Le, 2019). These are two of the most common methods of improving model accuracy but were widely misunderstood by many at the time. Older CNN models had to choose to scale up one of the following dimensions to improve accuracy:

- a. **Width:** This refers to the number of channels that each layer of the neural network has. Increasing layer channels gives the model more accurate details and provides more opportunities to capture complex shapes and patterns that would otherwise be missed (Potrimba, 2024). However, this would also mean that the model had to take longer because it would have to apply more filters in each layer. Inversely, models that have less width would be faster but potentially miss necessary details,

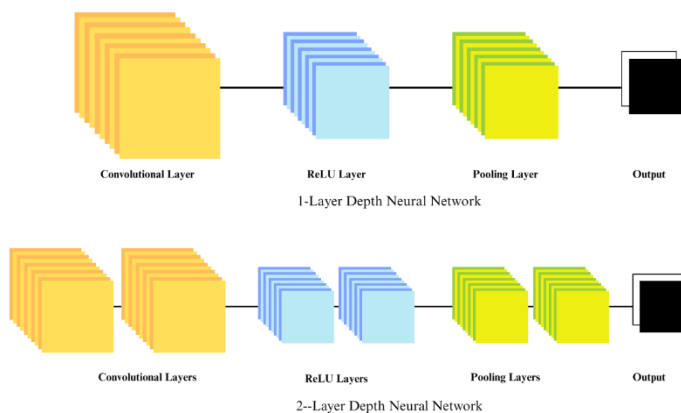
causing lower accuracy. This is the dimension that MobileNets (Howard, et al., 2017) scales in order to adjust to tasks meant for mobile devices.

Figure 5. Example of Different Layer Widths



- b. **Depth:** This refers to the actual number of layers in the network itself. Just like with width, having more layers in a neural network could lead to more intricate data representation and interpretation, but would also be much more resource-intensive (Potrimba, 2024). This is the principle that ResNets (He et al., 2016) uses--where the number in the model names indicate the depth (number of layers) that the model has.

Figure 6. Example of Different Layer Widths



- c. **Resolution:** This refers to the dimensions of the images fed into the model. This means scaling an image in terms of width and height, which increases the pixels it

contains and could provide more detailed information by allowing more kernels to be applied to an input image.

In other CNN models, these three dimensions are scaled arbitrarily; meaning that model width and depth, as well as image resolution are manually fine-tuned separately, leading to a lot of trial-and-error and potentially causing worse efficiency and accuracy. This is what EfficientNet's novel compound scaling method solves.

Compound scaling starts with a baseline model with a set depth d , width w , and resolution r . Then, a compound coefficient ϕ is defined, which dictates how much to scale each of these dimensions to (Potrimba, 2024). Each dimension's scaling factor is then derived from this compound coefficient:

$$\text{Width } (w) = \beta^\phi$$

$$\text{Depth } (d) = \alpha^\phi$$

$$\text{Resolution } (r) = \gamma^\phi$$

where β , α , and γ are constants that were determined by a small grid search, used to identify what the optimal baseline for each dimension is. Tan and Le reportedly found that the floating point operations (FLOPs), which is the amount of addition and multiplication required to process an input in a network, of each Convolutional Operation is proportional to d , w^2 , r^2 ; meaning doubling network depth will increase FLOPs by two, but doubling width or resolution will increase FLOPs by four. Using a small grid search, they found that the best values for each dimension's constants of their baseline model were $\alpha = 1.2$, $\beta = 1.1$, and $\gamma = 1.15$, under the constraint of $\alpha \cdot \beta^2 \cdot \gamma^2 \approx 2$. Their baseline model, EfficientNet-B0, uses these constants as the actual dimensions of the machine, having a coefficient of 0. Then, every succeeding variant from B1 to B7 applies a larger compound coefficient, uniformly scaling all the important dimensions to ensure minimum FLOPs relative to the accuracy of the results.

The body of EfficientNet-B0 as well as B1-B7 is based on MobileNet V2's architecture which "uses Mobile Inverted Convolutional (MBConv) Layers" (Potrimba, 2024), which combines depth-wise separable convolutions with inverted residual blocks. These MBConv layers each have different configurations such as kernel (filter) size, stride, and SE ratio. They also have depth-wise separable convolutions, which are split into depth-wise convolutions and pointwise convolutions; and squeeze-and-excitation layers which helps the model to require fewer computations and parameters.

Each model was trained on ImageNet, a large-scale image database used as a comparative baseline for performance evaluation between CNNs. In their study, Tan and Le compared the EfficientNet variants with different MobileNets and ResNets, grouping the ones with similar top-1/top-5 accuracies. They then compared the Parameters and FLOPs of the models in each group relative to the Parameters and Flops of the EfficientNet model in their group. The results in each group showed the EfficientNet models achieving similar and sometimes even better accuracy scores than the other models while using a fraction of the resources. Most notably, the B7 model achieved an equal performance with the best GPipe (Huang et al., 2018) model at the time, while being 8.4x smaller than it.

Figure 7. EfficientNet Performance on ImageNet (Tan and Le, 2019).

Model	Top-1 Acc.	Top-5 Acc.	#Params	Ratio-to-EfficientNet	#FLOPs	Ratio-to-EfficientNet
EfficientNet-B0	77.1%	93.3%	5.3M	1x	0.39B	1x
ResNet-50 (He et al., 2016)	76.0%	93.0%	26M	4.9x	4.1B	11x
DenseNet-169 (Huang et al., 2017)	76.2%	93.2%	14M	2.6x	3.5B	8.9x
EfficientNet-B1	79.1%	94.4%	7.8M	1x	0.70B	1x
ResNet-152 (He et al., 2016)	77.8%	93.8%	60M	7.6x	11B	16x
DenseNet-264 (Huang et al., 2017)	77.9%	93.9%	34M	4.3x	6.0B	8.6x
Inception-v3 (Szegedy et al., 2016)	78.8%	94.4%	24M	3.0x	5.7B	8.1x
Xception (Chollet, 2017)	79.0%	94.5%	23M	3.0x	8.4B	12x
EfficientNet-B2	80.1%	94.9%	9.2M	1x	1.0B	1x
Inception-v4 (Szegedy et al., 2017)	80.0%	95.0%	48M	5.2x	13B	13x
Inception-resnet-v2 (Szegedy et al., 2017)	80.1%	95.1%	56M	6.1x	13B	13x
EfficientNet-B3	81.6%	95.7%	12M	1x	1.8B	1x
ResNeXt-101 (Xie et al., 2017)	80.9%	95.6%	84M	7.0x	32B	18x
PolyNet (Zhang et al., 2017)	81.3%	95.8%	92M	7.7x	35B	19x
EfficientNet-B4	82.9%	96.4%	19M	1x	4.2B	1x
SENet (Hu et al., 2018)	82.7%	96.2%	146M	7.7x	42B	10x
NASNet-A (Zoph et al., 2018)	82.7%	96.2%	89M	4.7x	24B	5.7x
AmoebaNet-A (Real et al., 2019)	82.8%	96.1%	87M	4.6x	23B	5.5x
PNASNet (Liu et al., 2018)	82.9%	96.2%	86M	4.5x	23B	6.0x
EfficientNet-B5	83.6%	96.7%	30M	1x	9.9B	1x
AmoebaNet-C (Cubuk et al., 2019)	83.5%	96.5%	155M	5.2x	41B	4.1x
EfficientNet-B6	84.0%	96.8%	43M	1x	19B	1x
EfficientNet-B7	84.3%	97.0%	66M	1x	37B	1x
GPipe (Huang et al., 2018)	84.3%	97.0%	557M	8.4x	-	-

We omit ensemble and multi-crop models (Hu et al., 2018), or models pretrained on 3.5B Instagram images (Mahajan et al., 2018).

Despite every EfficientNet variant being much more efficient than the other models in their respective groups, choosing which one to use still depends on the trade-off between efficiency and accuracy necessary for a specific project. In the context of DR identification and classification, hospitals and clinics that specialize in ophthalmology would need fast results without sacrificing accuracy. Despite EfficientNet-B0 being ~7% less accurate than B7, the most accurate version, it is still 13x smaller and 95x less computationally expensive, making it the ideal model to use for these low-resource environments.

Metrics

When evaluating the performance of models, particularly in the context of diabetic retinopathy classification, choosing the right metrics is crucial. As different metrics highlight various aspects of a model's ability to classify correctly, ensure reliable disease detection, and balance computational efficiency.

1. Accuracy, F1-Score, Precision, and Recall

a. Accuracy

Accuracy is the metric that shows the percentage of right answers and provides a general measure of the quality of the model. However, accuracy alone may not be sufficient, as it may fail to capture important complexities. (Conciatori et al., 2024). Accuracy is a widely used metric in classification tasks to evaluate overall effectiveness and correctness. It measures how well the model classifies instances across the dataset. Additionally, it does not always account for deeper nuances like class imbalances or misclassification costs.

In medical applications, research suggests that an accuracy threshold of at least 90% is generally considered essential for models to be deemed dependable in clinical settings

(Kim, 2021). Falling below this benchmark poses challenges in ensuring the accuracy or precision needed to mitigate classification risks. Furthermore, classifying diabetic retinopathy, accuracy plays a significant role in determining the reliability of disease classification models, ensuring that predictions align with real clinical outcomes. In the sense that accuracy alone may not be sufficient, it must be interpreted alongside other metrics that consider false positives and false negatives, due to imbalanced datasets, where some severity levels may be underrepresented.

b. Precision

Precision measures the ratio of correctly predicted positive instances (true positives) to all instances predicted as positives (true positives and false positives) (Obi, 2023). It ensures that a model's positive classifications are highly reliable.

In the context of diabetic retinopathy classification, precision plays a crucial role in ensuring positive predictions. A high precision score indicates that when the model classifies a particular severity level of diabetic retinopathy, it is highly accurate, minimizing unnecessary diagnoses and avoiding unwarranted treatments.

c. Recall

Recall, or also known as sensitivity, measures the proportion of actual positive cases correctly identified by the model (Obi, 2023). In diabetic retinopathy, it reflects the model's ability to capture all true cases of diabetic retinopathy. Recall is vital in detecting as many true cases as possible.

A high recall score means that the model effectively identifies individuals with diabetic retinopathy, minimizing the likelihood of misdiagnoses.

d. F1-Score

F1-Score is the harmonic mean of precision and recall, balancing both metrics to provide a more comprehensive evaluation. It is useful when both false positives and false negatives need to be considered (Obi, 2023). As a key performance measure, it plays a crucial role in models handling imbalanced datasets, ensuring that misclassification rates are properly managed and that both types of errors are accounted for effectively.

In diabetic retinopathy classification, F1-Score helps maintain a balanced approach between precision and recall, ensuring that the model effectively captures true cases without excessively misclassifying others. A high F1-score indicates strong predictive reliability, making it an essential metric for evaluating overall model performance.

2. FLOPs (Floating Point Operations Per Second)

FLOPs quantify a model's computational workload by measuring the number of floating-point operations performed during a single forward pass (Chen et al., 2023). This metric is essential for assessing model efficiency, comparing architectures, and determining hardware suitability, particularly across diverse platforms. In medical applications, FLOPs play an important role in optimizing models for disease classification, ensuring that computational demands align with real-world deployment constraints such as speed and accessibility.

When classifying diabetic retinopathy, FLOPs determine how efficiently a model processes, balancing computational demands with real-world usability. Models with optimized FLOPs can achieve a high diagnostic accuracy, which is crucial for real-time screening in clinical settings. In the study, Detection of Diabetic Retinopathy using Convolutional Neural Networks for Feature Extraction and Classification (DRFEC) by Das

et al. (2023), FLOPs were used to evaluate the computational efficiency of 26 state-of-the-art CNN architectures, comparing their ability to process fundus images while maintaining high diagnostic accuracy. Optimized FLOPs ensured that models could balance complexity and inference speed. EfficientNetB4 emerged as the most efficient model, achieving high validation accuracy (79.11%) while maintaining a manageable computational workload.

FLOPs ensure that when evaluating computational efficiency, models remain scalable, deployable, and effective. By optimizing FLOPs, models can provide accurate predictions while maintaining feasibility for medical applications, particularly in classifying diabetic retinopathy.

In the study, Development and validation of predictive models for diabetic retinopathy using machine learning by Yang & Yang (2025), the metrics Accuracy, Recall, Precision and F1-Score were used to assess the effectiveness of multiple machine learning models for diabetic retinopathy classification. The experimental results demonstrated that Random Forest achieved an accuracy of 95.67%, precision of 95.89%, recall of 95.67%, and an F1-score of 0.957.

The selection and application of these metrics will allow us to properly evaluate reliability, accuracy, and computational efficiency of the enhanced EfficientNet-B0 model for diabetic retinopathy classification.

Synthesis of the Reviewed Literature and Studies

Diabetic Retinopathy (DR) classification is an important task in medical imaging, as early and accurate detection of DR can significantly reduce the risk of permanent vision loss. Unfortunately, access to timely diagnosis remains a challenge in underserved and marginalized communities, where medical professionals and diagnostic resources are

often scarce. This makes the development of automated and accessible DR screening tools more urgent.

A key difficulty in DR classification lies in the nature of the lesions themselves. Most clinically relevant features—such as microaneurysms, hemorrhages, and cotton wool spots—are small, subtle, and fine-grained. These characteristics make detection inherently challenging for conventional Convolutional Neural Networks (CNNs), which are limited by their localized receptive fields and inductive biases. As reflected in the reviewed literature, many CNN-based models fail to achieve the level of accuracy necessary for clinical deployment, especially in detecting early-stage DR.

While several studies have explored the use of preprocessing techniques and data augmentation strategies to improve CNN performance, the gains achieved through these methods alone remain insufficient. This shortfall underscores the need to pursue more fundamental architectural improvements.

Attention mechanisms, particularly self-attention, have emerged as a powerful alternative due to their ability to model long-range dependencies and focus on informative regions in the input. Vision Transformers (ViTs), which leverage these mechanisms, have shown promising results in DR classification tasks. However, their high computational requirements pose significant challenges for deployment in resource-constrained environments, an essential consideration in medical settings.

Among lightweight attention modules, Squeeze-and-Excitation (SE) and Convolutional Block Attention Module (CBAM) have gained traction for their efficiency. Yet even these mechanisms have not consistently delivered sufficient improvements to overcome the performance–efficiency tradeoff.

This growing gap between accuracy and efficiency highlights the need to explore other lightweight attention mechanisms that can provide performance gains without incurring substantial computational costs. It is within this context that the present study is situated, as we aim to explore attention mechanisms and integrate it into a DR classification model to make it not only accurate but also computationally efficient and suitable for deployment in low-resource healthcare environments.

Chapter 3

METHODOLOGY

Research Design

The study will employ a quantitative, quasi-experimental research design to evaluate the performance of the EfficientNet-B0 architecture with LSTL for Diabetic Retinopathy classification. The primary objective is to assess its effectiveness when compared to the baseline EfficientNet-B0 and the EfficientNet-B0 with CBAM model in terms of performance and efficiency.

Specifically, the core methodological approach will involve a model-level comparative analysis to evaluate the three (3) model configurations:

1. The baseline EfficientNet-B0 model with no attention mechanism,
2. EfficientNet-B0 augmented with CBAM, and
3. EfficientNet-B0 integrated with the LSTL module

To ensure robust evaluation, a standardized dataset—the APTOS 2019 Blindness Detection dataset—will be utilized across all system configurations to ensure comparability. The methods, k-fold cross-validation and Friedman testing, will be used to evaluate the models in various performance and efficiency metrics, specifically:

- Accuracy
- Precision
- Recall
- F1-score
- and FLOPs as an efficiency metric.

As the independent variable in this study is the attention mechanism applied to the model (none, CBAM, or LSTL), and the dependent variables are the performance and efficiency outcomes measured after training (accuracy, precision, recall, F1-score, and FLOPs), this ensures that the design is aligned with the study's stated problems and the dependent and independent variable relationships in the conceptual framework. Furthermore, the controlled setup ensures that any observed differences in performance can be attributed to the integration of attention-mechanisms explored alone.

Sources of Data

The dataset that will be used in this study is the APTOS 2019 Blindness Detection dataset, which contains 3662 high-resolution retinal fundus images, each labeled according to the severity of Diabetic Retinopathy (DR). The classification follows a five-grade scale: 0 (No DR), 1 (Mild), 2 (Moderate), 3 (Severe), and 4 (Proliferative DR). Each image is linked to a label provided in an accompanying CSV file.

The images in the dataset vary in resolution, lighting conditions, and visual clarity, closely reflecting the variability encountered in real-world clinical settings. This makes the dataset particularly suitable for evaluating the robustness of DR classification models. Originally released as part of a machine learning competition organized by the Asia Pacific Tele-Ophthalmology Society, it remains a widely used benchmark in medical image classification research.

Research Instrument

The researchers will utilize the EfficientNet-B0 Convolutional Neural Network integrated with an LSTL Module attention mechanism to develop a model for classifying DR from retinal fundus images. These images will be sourced from the publicly available APTOS 2019 Blindness Detection dataset, which contains pre-labeled high-resolution

images categorized into five DR severity levels. The selected images will be used for both training and testing the model.

Experiment Paper

To gather the necessary data for answering the research questions, two experiment papers will be used to evaluate the performance of the proposed EfficientNet-B0 with LSTL against the two other configurations. Experiment paper 1 will measure and compare each model's performance based on five evaluation metrics: Accuracy, Precision, Recall, F1 Score, and FLOPs. Experiment paper 2 will be used for the comparison of the three models statistically. Both experiment papers aim to help answer the research questions posed in the study.

Software/Hardware Tools

In this study, the researchers will use the following software on a laptop with the following minimum specifications: Windows 11 64-bit OS, GPU Support, and 16GB of RAM, to develop and evaluate a deep learning model for classifying diabetic retinopathy from retinal fundus images:

Python is a high-level, general-purpose programming language widely used in data science, machine learning, and artificial intelligence. In this study, the researchers will use Python as the core programming language for building the deep learning model used to classify diabetic retinopathy from retinal fundus images.

PyTorch is an open-source deep learning python library that provides tools for developing and training neural network models. Its flexibility and GPU acceleration support make it suitable for computer vision tasks. The researchers

will use PyTorch to construct and train the EfficientNet-B0 model integrated with a Local-Global Spatial Transformer Layer (LSTL) attention mechanism.

NumPy is a fundamental Python library for numerical computing. It enables efficient handling of multi-dimensional arrays and matrices. The researchers will use NumPy for processing image data and performing essential mathematical operations during model development.

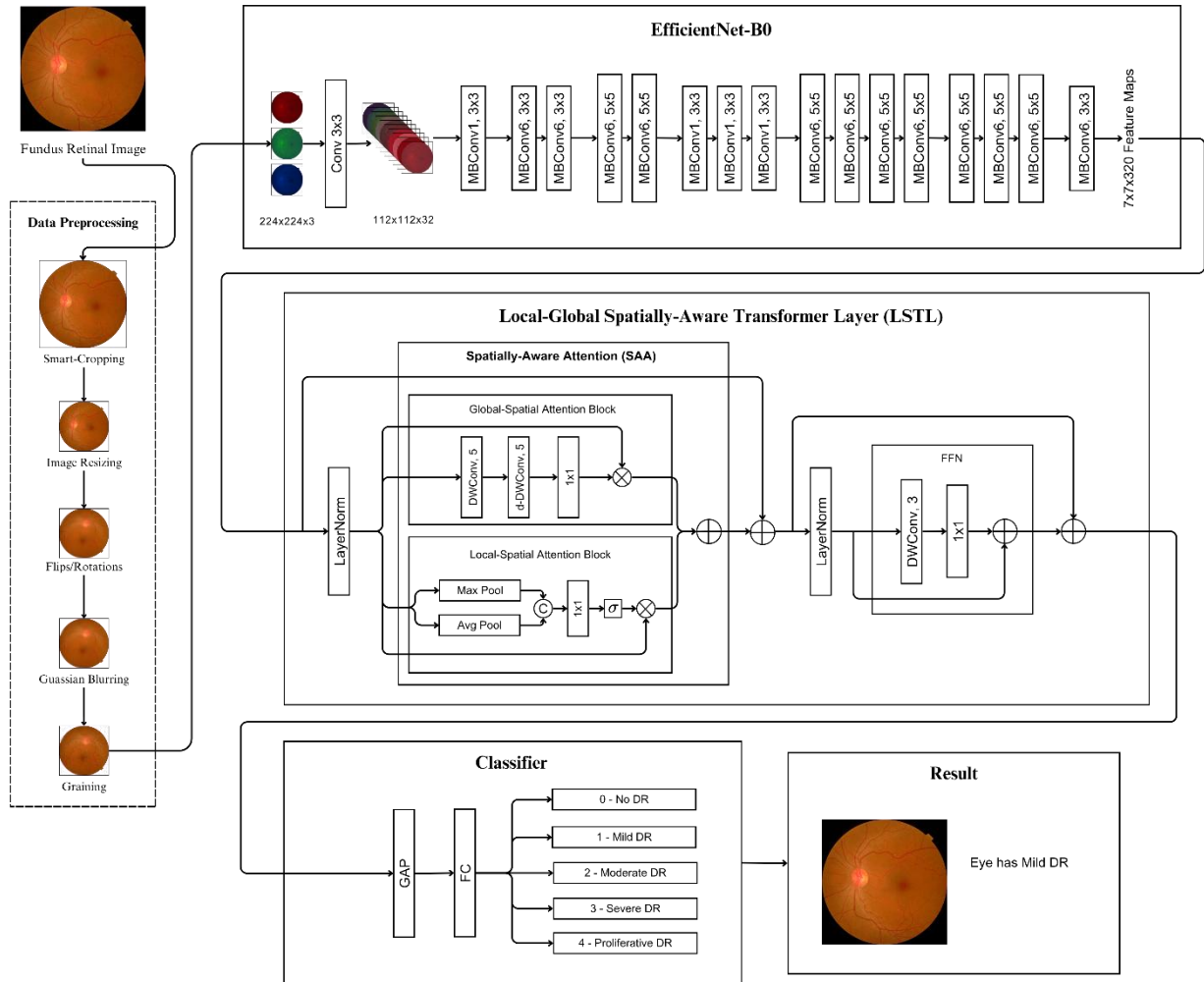
Pandas is a Python library used for data manipulation and analysis. It offers data structures such as DataFrames for handling structured datasets. In this study, Pandas will be used to load and organize image labels from the APTOS 2019 dataset's CSV file.

Matplotlib is a Python library for creating static and interactive visualizations. The researchers will use Matplotlib to visualize training performance metrics, including accuracy, data loss, and AUC curves, helping to evaluate and improve the model's behavior.

Scikit-Learn is a comprehensive and widely used machine learning library. It offers powerful and convenient tools for evaluating the model's performance. This library will provide key functions to gather the performance metrics of each of the classes to be used for the statistical analysis.

System Architecture

Figure 8. Overall System Architecture of the Study

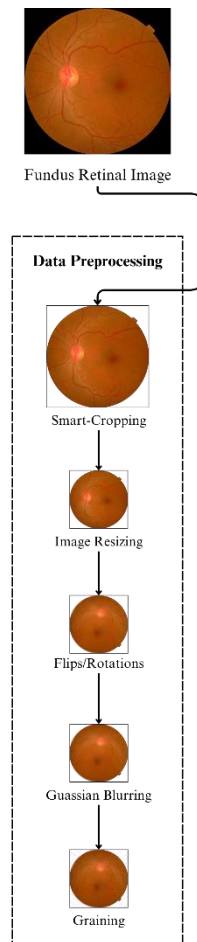


Preprocessing

The preprocessing stage will start with smart cropping, which will isolate the retina from the black peripheral background that accompanies the dataset. This will ensure that irrelevant areas will not dilute the feature learning process. After cropping, the images will then be resized to a uniform dimension of 224×224 pixels to meet the input requirements of the EfficientNet-B0 model. This will also

standardize spatial dimensions across the dataset for efficient batch processing. Finally, the resized images will be rotated over a random range (0.2 - 1.0) in either direction, to help produce variations of input images for each class.

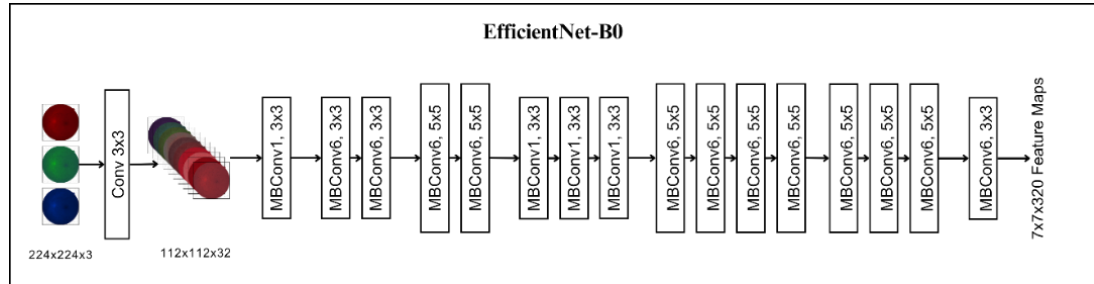
Figure 9. System Architecture on Preprocessing



Augmentation will also be done through random flips, rotations, gaussian blurring, and graining. These are meant to produce variations of the retinal images that reflect real-world variance when gathering these types of retinal scans.

EfficientNet-B0 for Feature Extraction

Figure 10. System Architecture for EfficientNet-B0 Backbone

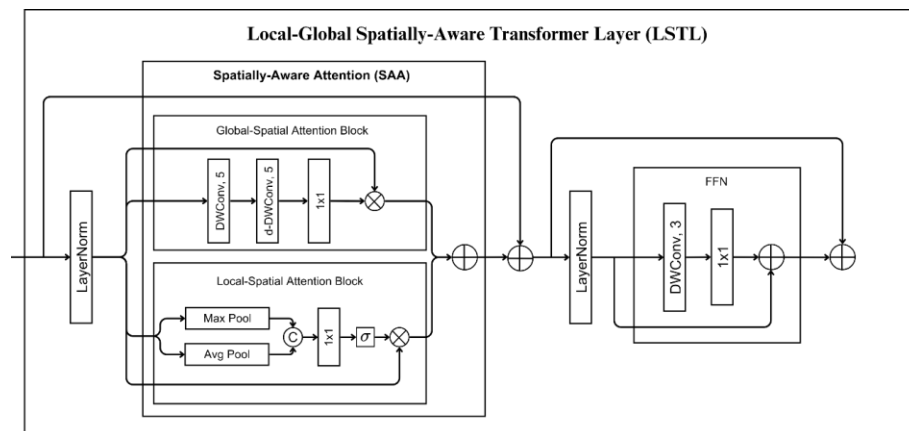


After preprocessing, the image will be passed through the EfficientNet-B0 backbone. This is so the image will be decomposed into progressively more abstract features through the MBConv blocks. The model has a total of 7 MBConv blocks, each meant to further reduce the outlying features and highlight the significant portion of each feature map.

These transformations are meant to condense the input image into a 7x7x320 feature map, which will distill the most meaningful patterns from the image. This model's strength lies in how it captures these multi-scale patterns while minimizing computational overhead, which is especially important in medical imaging where fast, reliable results are essential.

Local-Global Spatially-Aware Transformer Layer (LSTL)

Figure 11. System Architecture for LSTL



The extracted features will then be passed through the Local-Global Spatially-Aware Transformer Layer. However, before any attention is applied, the feature maps extracted will first undergo LayerNorm. This step is critical for stabilizing and accelerating the training process. Following normalization, the features will then be passed into the Spatially-Aware Attention mechanism, the core of LSTL. This mechanism is divided into two attention blocks: the Global Spatial Attention Block and the Local Spatial Attention Block, as illustrated in Figure 11.

The Global-Spatial Attention Block uses Depthwise Convolutions (DWConv) and dilated-Depthwise Convolutions (d-DWConv) to evaluate patterns across the entire retina, detecting changes in vessel shape and widespread lesions. The Depthwise convolutions have a kernel size of 5 and is meant to normalize each channel using a separate convolution layer for each one. It will then pass to a dilated-DWConv with the same kernel size of 5 and applies a dilation rate of 3. This dilation is to expand the receptive field of extractable features from the layer without increasing any of the parameters and computational cost, as it only separates each channel layer with its dilation rate. This dilation inserts gaps between the kernel elements, meaning each convolution will be able to capture more details in a wider area. The channel features will then pass through a 1x1 pointwise convolutional layer to combine all the information across the channels. Finally, the output of these three convolutions will be multiplied element-wise with the original 7x7x320 feature map to weigh the importance of each channel, which would indicate significance in terms of detecting a specific feature.

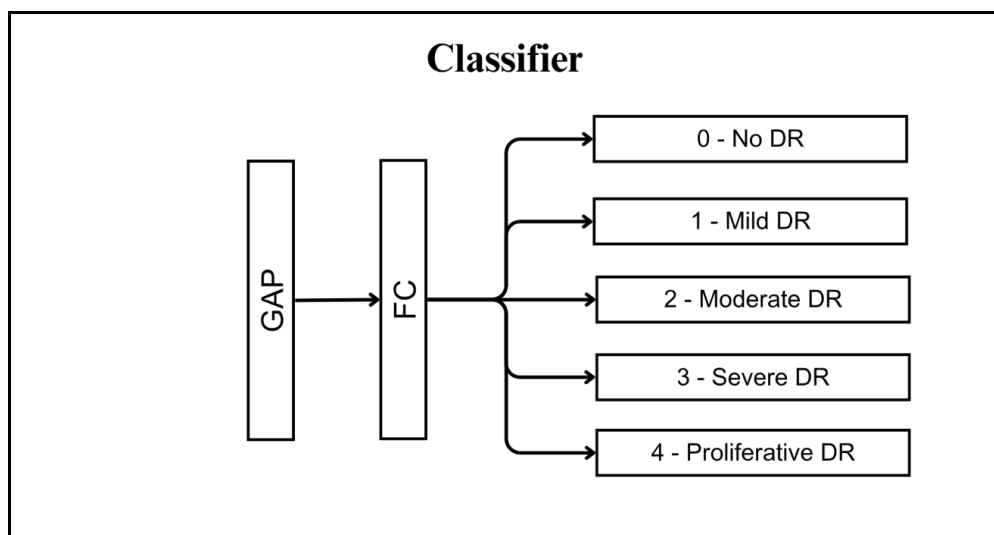
Simultaneously, the Local-Spatial Attention Block will employ max and average pooling to capture different aspects of local information. It will take in the

7x7x320 feature map from the EfficientNet-B0 backbone as well. the LSAB has a max-pooling and an average pooling operation. Max pooling is meant to extract the highest pixel value from each patch of the feature map. This means this operation will select the most significant pixel for each channel, ensuring that only the most important features get extracted. Average pooling simultaneously works along global pooling, where it will calculate the mean from each patch of the feature map. This helps get a more generalized representation of features from local regions. The outputs of both these operations will then get concatenated along channel dimensions, combining both prominent and generalized local spatial information. This output will also get passed through a 1x1 pointwise convolution to allow the model to learn the weighted combination of the prominent and generalized local features that will be extracted by the pooling layers. The result will then pass through a sigmoid activation function to flatten the feature maps between 0 and 1, where 1 would indicate higher importance and 0 would indicate least importance. Just like GSAB, the result of this block will be multiplied with the original 7x7x320 feature map to apply the weights of these extracted local features.

This dual attention mechanism will allow the model to assess each region in the context of the whole image, which is a crucial capability for differentiating between DR stages. The outputs of these two blocks will be combined through element-wise addition and then passed to subsequent layers. These refined features will once again go through Layer normalization and pass through an FFN. This is added mostly to further process the feature maps and to contribute to the extraction of salient features from key lesion areas.

Classification Layer

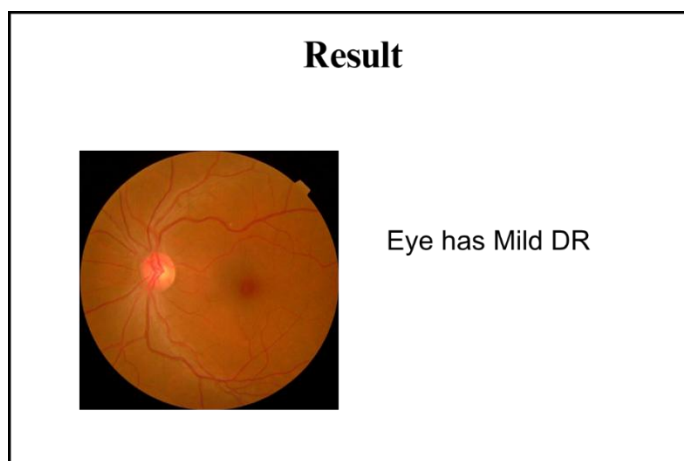
Figure 12. System Architecture for the Classifier Layer



The outputs are then to be passed through a Global Average Pooling (GAP) layer, which compresses spatial information by averaging each feature channel into a single value. Afterwards, the pooled features will be passed into a Fully Connected layer, which will translate the information into a prediction between five DR Severities: either 0 (No DR), 1 (Mild), 2 (Moderate), 3 (Severe), or 4 (Proliferative).

Model Output

Figure 13. System Architecture of the Classification Result



The predicted DR stage will be translated into a diagnostic result after classification. For example, if the model detects minor signs like isolated microaneurysms, it will output “Eye has Mild DR.” The result is presented alongside the original retinal image for context, offering a visual reference that supports clinical review. The clarity and precision of this final stage is essential, as it determines the reliability and integrity of the entire diagnostic pipeline in real-world clinical settings.

Data Gathering/Data Generation Procedure

APTOS’ 2019 BD dataset was gathered and prepared by the Aravind Eye Hospital in India (Herrero, M. 2019). According to their description, the fundus photographs were collected in various conditions and environments over a long period of time. The gathered samples were then reviewed and labeled by a group of trained doctors, following the principle of the International Clinical Diabetic Retinopathy Disease Severity Scale (ICDRSS). Consequently, they devised the scaling system between the 5 categories which this dataset uses.

The following data preprocessing steps are designed to prepare the images for classification analysis while retaining characteristics that reflect real-world implementation. The study will loosely follow a pre-processing method based on Duc Nguyen’s implementation for a ResNet50 baseline (Ratthachat, 2019):

- 1. Loading in the training/test data frames** - The fundus images and their corresponding labels from the CSV file will be loaded into appropriate data structures for processing.

2. **Smart Cropping** - Numpy's auto-crop feature will be used to crop out the black border present in each image by making it delete all the pixels which have a value of 0.
3. **Resizing** - Every cropped image will be normalized to a 224x224 size, ensuring equal dimensionality across the dataset and consistent batch processing for when model training begins. This aligns with EfficientNet's compound scaling, as the baseline model is required to have this resolution. This also makes it so that every image will be spherical, better emulating real-life retinas.

The purpose of doing minimal preprocessing is to be able to train the model using data that would accurately reflect real-life implementation of the proposed neural network.

To enhance model robustness and generalization to varying image qualities, on-the-fly image augmentation will be applied to the training set. This will involve randomly rotating images within a 20%-100% range, the random application of Gaussian blur with varying σ values (0.5 to 2.0), and the addition of digital grain/noise (25%, 40%, 50%). This approach will increase the diversity of training samples presented to the model without altering the size of the original dataset.

Data Sampling for Model Evaluation

To ensure robust and reliable performance evaluation, this study will utilize a k-fold stratified cross-validation scheme instead of a single train/test split. To account for computational resources and dataset size, as well as the time given to conduct the study, the most practical k value to use is 5. This means that each fold would have approximately $3662/5 \approx 732$ test images. The training set would be $3662 \times 0.8 \approx 2930$ images. This provides a substantial training set in each fold.

In each fold of the cross-validation:

1. The complete dataset will be partitioned into k mutually exclusive and collectively exhaustive folds.
2. In each iteration, one fold will be designated as the validation/test set, and the remaining 4 folds will be combined to form the training set.
3. This process will be repeated 5 times, ensuring that each image in the dataset serves as part of the validation/test set exactly once.
4. To maintain class distribution consistency across folds, stratified k -fold cross-validation will be implemented, ensuring that the proportion of each DR severity level (0 to 4) is approximately preserved in both training and validation sets for every fold.

This cross-validation approach will be applied to all three of the models. This is to allow for training and evaluation multiple times on different subsets of the data, providing more reliable performance metrics and enabling the appropriate statistical analyses for it.

Ethical Considerations

This study will ensure ethical compliance by exclusively using the APTOS 2019 Blindness Detection dataset, a publicly available and anonymized dataset. All retinal fundus images are devoid of personally identifiable information and were originally released for research and educational purposes. The dataset will be accessed and utilized in accordance with the usage rights and licensing conditions provided by APTOS.

No direct human participation or additional data collection will be involved in this research. Furthermore, the deep learning models developed in this study will be used solely for academic and research purposes and will not be deployed in clinical or diagnostic settings.

Data Analysis

After training and testing, the necessary metrics will be measured in line with the study's purpose. To evaluate the enhanced EfficientNet-B0 model's performance, various data must first be measured. Namely, these are:

Confusion Matrix

A confusion matrix will be used to represent the classification results of each class to help see the types of errors that the model will make. Since there are 5 classes (No DR, Mild DR, Moderate DR, Severe DR, and Proliferative DR), a 5x5 matrix will be used. This will help illustrate the distribution of **True Positives (TR)**, **True Negatives (TN)**, **False Positives (FP)**, and **False Negatives (FN)** that the model will make.

Figure 14. Confusion Matrix of Classification Results

True Classification	0	0	0	0	0	
	1	0	0	0	0	
	2	0	0	0	0	
	3	0	0	0	0	
	4	0	0	0	0	
		0	1	2	3	4
		Predicted Classification				

Performance Metrics

After getting the distribution of the classification results (TP, FP, TN, FN), the Accuracy, Precision, Recall, and F1-Score for each class per fold will be

calculated to get the overall performance of the model. This is adapted from Aasrith's (2024) measurement method for measuring their model's overall accuracy in classifying DR.

- a. **Accuracy** - This is the proportion of correct DR classifications among the k-fold dataset that the models will make. Each model's overall accuracy will be measured per fold and averaged to get their total accuracy. This will serve as the primary metric for determining each model's correctness.

$$Acc = \frac{TP + TN}{Total\ Dataset}$$

- b. **Precision** - This is the proportion of correctly classifying a DR class among all cases to be predicted as that specific class. This is a metric that will be measured multiple times, once for each class before being averaged to get the overall precision of each fold. A high precision score would mean fewer false classifications of each DR class.

$$Precision = \frac{TP}{TP + FP}$$

- c. **Recall** - This is the proportion of the actual DR Classes that will be correctly classified over what the model classified. This is a metric that will also be measured multiple times, once for each class before being averaged for each fold's overall recall. Higher recall would mean fewer false negatives.

$$Recall = \frac{TP}{(TP + FN)}$$

- d. **F1-Score** - This is the harmonic mean of the Precision and Recall scores. It will provide a single metric to help balance both scores and serve as another metric of actual accuracy. This will be calculated for each class per fold and will

serve as a more robust metric for accuracy that disregards variance between the number of classes.

$$F1 - Score = 2 * \frac{(Precision * Recall)}{(Precision + Recall)}$$

Computational Cost Metrics

Another metric that must be measured is the computational cost of the model to determine how much computational overhead increases relative to the increase in accuracy. For this, **Floating Point Operations** will be used as the metrics. This will be based on Babita & Nayak's 2024 study, which used this metric to compare their attention module network to other CNNs and Vision Transformers.

Floating Point Operations (FLOPs)

This will measure the number of operations required for a single forward pass through the network. This measurement is used by neural networks as a representation for computational intensity. The `pytorch_model_summary` library will be used to calculate this accurately.

Statistical Analysis

To evaluate the performance of the EfficientNet-B0 enhanced with LSTL against the baseline EfficientNet-B0 model and EfficientNet-B0 with CBAM, a statistical analysis will be conducted using k-fold cross-validation and the Friedman Test. This approach ensures a more reliable and generalizable assessment of model performance compared to a single train/test split.

Performance Metrics Analysis: Accuracy, Precision, Recall, and F1-Score

For each performance metric, the following statistical procedure will be applied:

1. **Data Acquisition from Cross-Validation** - From the 5-fold cross-validation, five distinct scores will be obtained for each performance metric of each of the three models. These sets of scores are inherently related, as each model is evaluated on the same test fold in every iteration. The total scores of the models' metrics for each fold will be listed for comparison.

Table 1. Accuracy Comparison Table between EfficientNet-B0 with LSTL, EfficientNet-B0, and EfficientNet-B0 with CBAM

Accuracy Scores			
Fold (k)	EfficientNet-B0 + LSTL	EfficientNet-B0	EfficientNet-B0 + CBAM
1			
2			
3			
4			
5			

After listing the scores, the models will be ranked based on the highest scores for each fold from first, second, then third. In cases where two or more models achieve identical performance scores within a fold, they will be assigned the average of the ranks they would have collectively occupied. If two models tied for the second-best performance in a fold, they would both be assigned a rank of $(2+3)/2=2.5$. This process will be done for every performance metric.

Table 2. Accuracy Ranking Table between EfficientNet-B0 with LSTL, EfficientNet-B0, and EfficientNet-B0 with CBAM

Accuracy Ranking			
Fold (k)	EfficientNet-B0 + LSTL	EfficientNet-B0	EfficientNet-B0 + CBAM
1			
2			

3			
4			
5			
Total			

2. **Overall Comparison using Friedman Test** - To determine if there is an overall significant difference in performance among the three models for each performance metric, the Friedman Test will be employed. The Friedman test computes a chi-squared statistic based on the ranks of the performances within each fold. A low p-value would indicate statistical evidence to reject the null hypothesis, suggesting that a significant difference exists somewhere among the model performances.

Friedman Test Formula

$$X_F^2 = \frac{12}{nk(k+1)} \sum_{j=1}^k R_j^2 - 3n(k+1)$$

Where:

n = number of folds

k = number of models

R_j = the sum of the ranks for the j-th model across all n folds.

The result of this test will then be compared to a chi-squared distribution with k-1 degrees of freedom to get its p-value. This p-value will represent the probability of observing a test statistic as extreme as, or more extreme than the calculated X_F^2 , assuming the null hypothesis is true. A significance level of $\alpha=0.05$ will also be set. This means there is a 5% risk of incorrectly rejecting the null hypothesis (Type I error) when it is actually true.

3. **Post-Hoc Analysis for Pairwise Comparisons (Nemenyi Test)** - If the Friedman Test yields a statistically significant result for a metric ($p < 0.05$), it indicates that a difference exists, but it does not specify *which* particular model pairs are significantly different. To address the research objectives, specifically: to prove LSTL's improvement over the Baseline model and over CBAM, a post-hoc test is necessary.

The Nemenyi post-hoc test will be used to determine if the difference between the **average ranks** of any two models is large enough to be considered statistically significant. If the absolute difference between the average ranks of two models exceeds the Critical Difference (CD), then those two models will be considered significantly different.

Critical Defense Formula

$$CD = q_{\alpha, k, \infty} \sqrt{\frac{k(k+1)}{12n}}$$

Where:

α : significance level (0.05)

k: The number of models being compared

∞ : The degrees of freedom for the studentized range distribution

n: The number of folds (5)

The post-hoc analysis will specifically focus on two crucial pairwise comparisons for each performance metric:

- **Comparison 1 - LSTL Variant vs. Baseline EfficientNet-B0:** This comparison aims to determine if the EfficientNet-B0 with LSTL model significantly improves upon the Baseline EfficientNet-B0 model. A statistically significant result ($p < 0.05$)

after correction) for this pair, where the LSTL variant's median performance is higher, would support the alternative hypothesis for the specific metric.

- **Comparison 2 - LSTL Variant vs. CBAM Variant:** This comparison aims to determine if the EfficientNet-B0 with LSTL model performs significantly better than the EfficientNet-B0 with CBAM model. A statistically significant result ($p < 0.05$ after correction) for this pair, showing higher median performance for the LSTL variant, would support the alternative hypothesis for each metric.

Computational Cost Analysis: FLOPs

Floating Point Operations (FLOPs) will be directly measured for each model using the `pytorch_model_summary` library. As FLOPs represent a deterministic measure of computational intensity for a given architecture and input size, no statistical tests are required for this metric. The FLOPs values for each model will be reported and qualitatively compared to assess the computational overhead associated with each variant relative to its performance and the statistical findings on accuracy and other metrics.

Interpretation of Results

If the Friedman test results are significant, the adjusted p-values from the Nemenyi post-hoc tests will be used to draw conclusions. A p-value less than $\alpha = 0.05$ for a specific pairwise comparison will indicate a statistically significant difference. The overall interpretation will synthesize the findings from all performance metrics and computational cost. For instance, if the LSTL variant demonstrates statistically significant improvements in key performance metrics compared to both the Baseline and CBAM models, while maintaining comparable or acceptable FLOPs, it would strongly support its efficacy for DR classification.

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APPENDICES

Appendix 1: Research Instrument

The table presented below will present the outcomes of the testing phase, where k-fold cross validation will be used for every 20% of the APTOS 2019 Blindness Detection dataset to compare the performance of EfficientNet-B0 and LSTL with the baseline EfficientNet-B0 and EfficientNet-B0 with CBAM. Each model will use its own set of tables. Each row will represent a specific test case, detailing the file name, the actual severity of diabetic retinopathy, and the predicted severity level. A checkmark (✓) will indicate the class predicted by the model for that instance. These results will help assess the classification performance of each model and identify correct and incorrect predictions.

Table 3. Test Result Table for Each Fold per model

Test	File Name	Actual Severity	Predicted DR Severity				
			No DR	Mild	Moderate	Severe	Proliferative DR
1							
2							
3							
...							
N							

Appendix 2: Experiment Paper for measuring Performance Metrics

Objective:

To collect and evaluate the classification performance of EfficientNet-B0-LSTL with baseline EfficientNet-B0 and EfficientNet-B0 with CBAM in identifying diabetic retinopathy severity from retinal fundus images, using Accuracy, Precision, Recall, F1 Score, and FLOPs as evaluation metrics.

Materials:

- EfficientNet-B0 and variants
- Metric results per fold of each model
- $Accuracy = \frac{(TP+TN)}{Total\ Dataset}$
- $Precision = \frac{TP}{(TP+FP)}$
- $Recall = \frac{TP}{(TP+FN)}$
- $F1 - Score = \frac{Precision * Recall}{Precision + Recall}$

Where:

- **TP** = True Positive
- **TN** = True Negative
- **FP** = False Positive
- **FN** = False Negative

Procedure:

1. Prepare the APTOS 2019 Blindness Detection dataset and input the testing images into each of the three model configurations: baseline EfficientNet-B0, EfficientNet-B0 with CBAM, and EfficientNet-B0 with the proposed LSTL module.
2. Run each model on the test data to classify the severity of diabetic retinopathy in the images into one of five categories: No DR, Mild, Moderate, Severe, or Proliferative DR.
3. Record the results in a separate table for each model, indicating the file name, actual label, and predicted class. A checkmark (✓) will be used to show the predicted class per image.
4. Evaluate the accuracy of each model by comparing its predictions against the ground truth labels. Compute the Accuracy, Precision, Recall, F1 Score, AUC, and FLOPs using a confusion matrix and appropriate metric formulas.

Table 4. Table for confusion matrix

Labeled DR Severity	Predicted DR Severity				
	No DR	Mild	Moderate	Severe	Proliferative DR
No DR					
Mild DR					
Moderate DR					
Severe DR					
Proliferative DR					

Table 5. Performance Evaluation Table in Recognizing Different DR Severity

DR Severity	Accuracy	Precision	Recall	F1-Score
No DR				
Mild DR				
Moderate DR				
Severe DR				
Proliferative DR				

Appendix 3: Experiment Paper for Statistical Analyses

Objective:

To answer the hypothesis and draw a statistical inference between EfficientNet-B0+LSTL and the other two EfficientNet-B0 variants in classifying diabetic retinopathy severity. This means using the gathered performance metric scores (Accuracy, Precision, Recall, F1-Score) and computational cost (FLOPs) and applying the appropriate statistical tests for them.

Materials:

- Performance evaluation results for each model across all 5 cross-validation folds generated in Appendix 2.
- Computational cost (FLOPs) for each model.

- Statistical software capable of performing Friedman's test and Nemenyi post-hoc comparisons.
- $X_F^2 = \frac{12}{nk(k+1)} \sum_{j=1}^k R_j^2 - 3n(k+1)$
- $CD = q_{\alpha, k, \infty} \sqrt{\frac{k(k+1)}{12n}}$

Procedure:

Five distinct performance scores for each metric will be obtained from each model from 5-fold cross-validation.

1. For each performance metric, models will be ranked from best (Rank 1) to worst (Rank 3) within each cross-validation fold. Tied scores will be assigned the average of their collective ranks.
2. The Friedman Test will be employed to determine if an overall significant difference exists in performance among the three models for each metric. This non-parametric test computes a chi-squared statistic based on the ranks, which will be used to find a p-value compared against $\alpha = 0.05$.
3. If the Friedman Test yields a significant result, the Nemenyi post-hoc test will be conducted to identify specific pairwise differences between models. This test calculates a Critical Difference (CD), and if the absolute difference in average ranks between two models exceeds the CD, they are considered significantly different.

Floating Point Operations (FLOPs) will be directly measured for each model using `pytorch_model_summary`. These deterministic values will be reported and qualitatively compared to assess computational overhead relative to performance.

Findings from all performance metrics and computational cost will be synthesized, interpreting significant Friedman results via Nemenyi test p-values to support claims of LSTL's efficacy or lack thereof compared to baseline and CBAM.

Table 6. Metric Comparison Table between EfficientNet-B0 with LSTL, EfficientNet-B0, and EfficientNet-B0 with CBAM per Fold

Metric Score			
Fold (k)	EfficientNet-B0 + LSTL	EfficientNet-B0	EfficientNet-B0 + CBAM
1			
2			
3			
4			
5			

Table 7. Metric Ranking Table between EfficientNet-B0 with LSTL, EfficientNet-B0, and EfficientNet-B0 with CBAM per Fold

Metric Ranking			
Fold (k)	EfficientNet-B0 + LSTL	EfficientNet-B0	EfficientNet-B0 + CBAM
1			
2			
3			
4			
5			
Total			

Appendix 4: System Mockup/Wireframe

